MAE 185 Final Project Report

Maelstrom

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Overview

This investigation evaluates how the sizing of fuel inlet and oxidizer vortex inlets impacts the internal flow characteristics of a vortex-cooled rocket engine configuration. Operating under the constraints of the original Maelstrom inputs, our work isolates the geometric effects of inlet diameters on internal chamber pressure, fluid velocity fields, outlet properties, and turbulence generation.

To conduct this study, our group altered the baseline CAD profile (8FL_Maelstrom_v6.step) into these scenarios:

	Δd_F **(Fuel inlet diameter change)**	**Δd_O** **(Oxidizer inlet diameter change)**	**Description**
Control	**0%**	**0%**	Established the baseline reference using a balanced entry area for both the axial fuel injector and the four tangential oxidizer ports.
Oversized	**+10%**	**+10%**	Expanded the inlet diameters of both the fuel injector and the oxidizer ports by 10% to study the effects of slower fluid velocity
Undersized	**-10%**	**-10%**	Constricted the inlet diameters of both the fuel injector and the oxidizer ports by 10% to study the effects of faster fluid velocity
High Shear	**+15%**	**-15%**	Increasing the fuel inlets and decreasing the oxidizer inlets by 15% to accelerate the vortex surrounding the fuel stream
Core Punch	**-15%**	**+15%**	Decreasing the fuel inlets and increasing the oxidizer inlets by 15% to accelerate the fuel injection through the center
Velocity Match	**0%**	**+13.4%**	Increasing oxidizer inlets to match the velocity of the fuel

Results Analyzed

We quantified and compared internal engine performance across all cases using the following metrics:

1. **Chamber Pressure:** Monitored along the central axis and upper injector face to trace core stagnation pressure levels.
2. **Chamber Velocity Contours:** Visual analysis of the chamber velocity fields
3. **Outlet Axial Velocity:** Mapped to track the boundary layer wrap of the outer helical vortex and its transition into the descending core vortex.
4. **Turbulence Kinetic Energy:** Quantified to evaluate shear-driven mixing layers where the high-speed tangential stream interacts with the central axial stream.

Introduction

Rocket engines operate in extremely demanding thermal and fluid-dynamic environments, where propellant injection and mixing play a critical role in the engine's performance. Rocket engines typically operate in 5 steps: propellant delivery, injection, ignition, combustion, and exhaust/thrust. Following the propellant delivery and injection, fuel and oxidizer must mix efficiently before the combustion stage to produce stable operation and effective thrust. As a result, injector geometry will influence flow velocity distribution, turbulence, recirculation, and mixing behaviors. Traditionally, methods such as regenerative, tangential, and radiative cooling are utilized in industry. However, these systems could increase the overall complexity, manufacturing requirements, and overall mass.

In this report, we review the applications of vortex chambers and their flow characteristics. A vortex chamber is a type of combustion chamber in which the oxidizer is injected to generate a swirling flow field, producing motion that enhances mixing and influences velocity, turbulence, and the temperature within the system. Previous investigations conducted by NASA researchers have found that the flow structure of a vortex chamber can enhance propellant mixing, which further promotes turbulence and creates distinct flow regions that reduce the heat transfer at the walls [1]. Project Maelstrom aims to investigate this application of such vortex chamber concepts to a student-scaled rocket engine. The design consists of a central axial fuel injector surrounded by four tangential oxidizer inlets.

The objective of this study is to examine how changing the fuel and oxidizer inlet radii will affect the flow characteristics of the Maelstrom rocket engine. Five configurations were analyzed in comparison to the original model: an oversized injector configuration, an undersized configuration, high shear, core punch, and velocity match. These simulations were performed using ANSYS Fluent to evaluate the chamber pressure, velocity distribution, outlet velocity, and turbulent kinetic energy. By comparing these configurations, the study seeks to determine how changing the size of the injector may influence the formation of the vortex and the flow behavior occurring inside the engine.

References

[1] Trinh, H. P., et al. *Evaluation of Vortex Chamber Concepts for Liquid Rocket Engines*. National Aeronautics and Space Administration, 2000. NASA Technical Report.

Methods

The Maelstrom engine was simulated using ANSYS Fluent under steady-state cold-flow conditions. Air was modeled as a compressible ideal gas, and the governing equations for the conservation of mass, momentum, and energy were solved using the finite volume method. Turbulence effects were modeled using the Realizable k-ε with enhanced wall treatment to capture turbulent mixing and swirling contours. The energy equation was enabled to account for temperature variations as the flow accelerates through the chamber and nozzle, while Sutherland's viscosity was used to represent the temperature dependence of air viscosity. These numerical models were used to evaluate vortex formations, velocity distributions, pressure, and turbulence with the following inputs:

$\dot{m}_{max,fuel} = 0.02376\ kg/s$

$\dot{m}_{max,ox} = 0.01188\ kg/s$

$OF = 1.8$

$p_{max} = 689.476 kPa$

$M_{nozzle} = 2.54$

With these numerical models and assumptions established, the geometry, mesh, and domain in ANSYS Fluent were completed using the following procedure:

Geometry Preparations

1. Set-up Workbench
    a. Open Ansys Workbench
    b. Drag and drop "Fluid Flow (Fluent): into the empty space
    c. In the box that shows up, right-click geometry
    d. Click Import Geometry and find the STEP file
    e. Right-click geometry again and click "Edit Geometry in SpaceClaim."
2. Parameterize
    a. Click the Design tab and select "Pull."
    b. Select the fuel hole surface and click the P
    c. Rename the parameter in the Groups tab to "Fuel_Diameter."
    d. Do the same thing for the four outlets by using the Ctrl key, and again rename the parameter
3. Extract the fluid volume
    a. Under the Prepare tab, click "Volume Extract."

b. Make sure it is on "Select Edges" and click on the edges for the fuel inlet, the four oxidizer inlets, and the bottom plenum hole
c. Click the green check mark
d. On the left structures tree, suppress all components that aren't the volume
    i. Click the check box next to each of the components
    ii. Right-click on each component and select "Suppress for Physics"

4. Create Named Sections
    a. Select the fuel inlet, then go to the Groups tab and select "Create NS" and name it fuel_inlet
    b. Do the same for the oxidizer inlets by hitting Ctrl and selecting all the inlets
    c. So the same for the plenum outlet
    d. Triple-click the volume to select all the walls, and hit the Ctrl key and deselect all the holes (fuel inlet, ox inlet, and plenum outlet)
5. Save your file

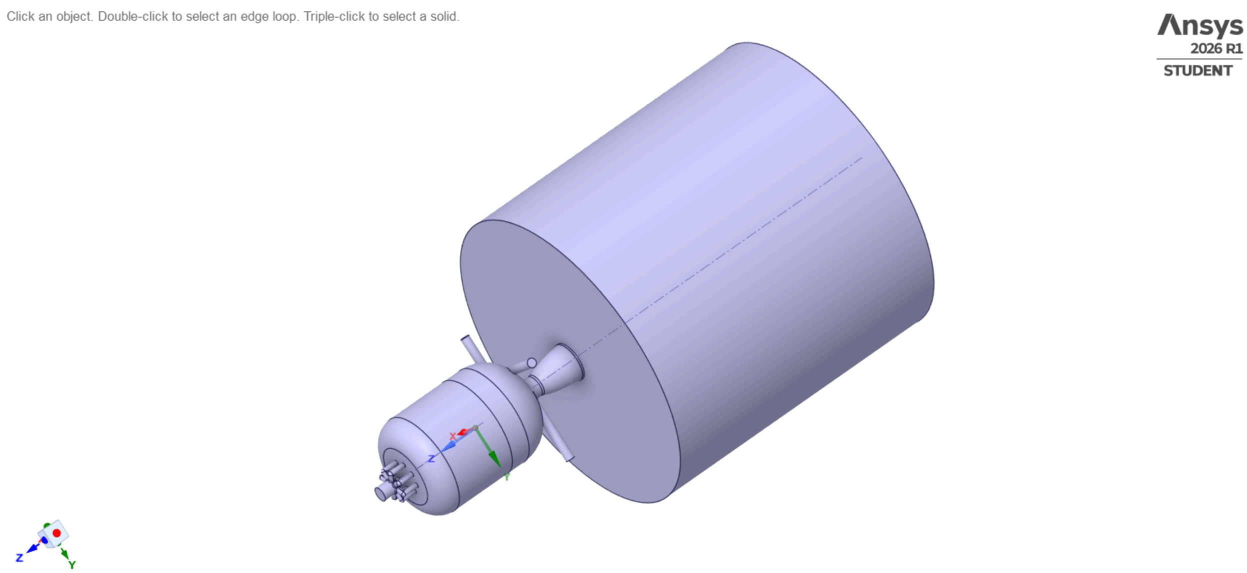


Figure 1: CAD model of the Maelstrom rocket engine after geometry preparations

Meshing

1. Launch Meshing Module
    a. After making sure to save your file, you can close SpaceClaim
    b. Go back to Ansys Workbench
    c. Double-click the mesh cell, and the meshing software should automatically open up
2. Set Global Parameters
    a. In the Outlines tree, select "Mesh."

b. Under Defaults, make sure Physics Preference is “CFD,” and Solver Preference is “Fluent.”
c. Under Sizing, make sure Capture Curvature is set to “Yes.”

3. Add Localized Face Sizing
    a. Right-click on Mesh in the Outline tree, then under Insert, hit Sizing
    b. In the details box, click the box next to Geometry
    c. Click the top fuel inlet and hit Ctrl to select the oxidizer inlets
    d. Click Apply, and the Geometry boxes should now say 5 faces
    e. Set the Element Size to 0.0005 m
4. Add Inflation Layers
    a. Right-click on Mesh in the Outline tree, then under Insert, hit Inflation
    b. For the Geometry, select the entire body and hit Apply
    c. Change the Boundary Scoping Method to Named Selections and in the Boundary box select “walls.”
    d. Ensure Maximum Layers is set to 5
5. Generate Mesh and Verify Quality
    a. Right-click Mesh in the Outline tree and select Generate Mesh
    b. Under Details of “Mesh,” expand the Quality section and change the Mesh Metric to Skewness, ensure the Max Skewness is below 0.95
    c. Save your mesh

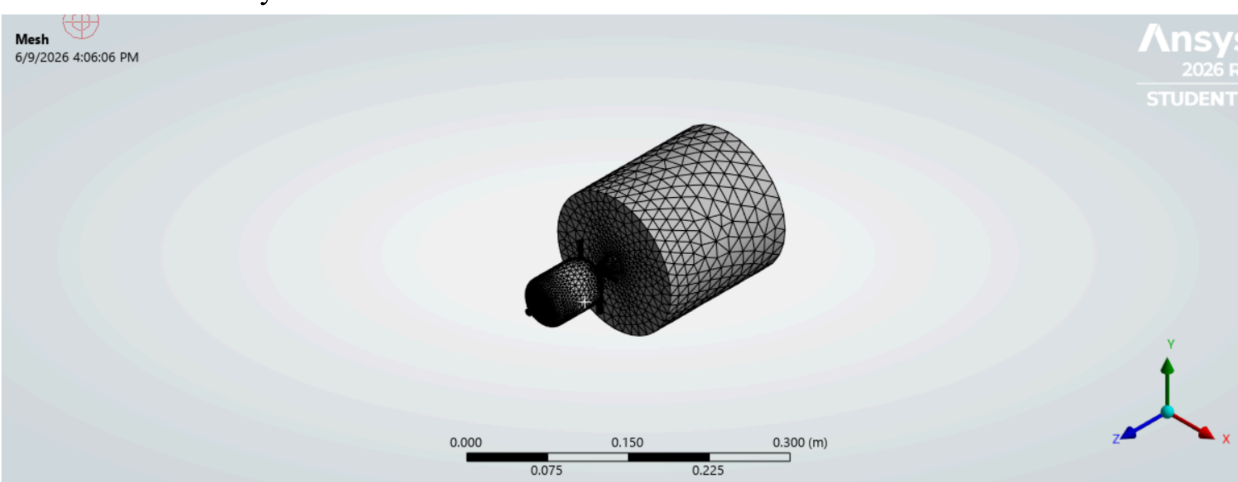


Figure 2: Computational mesh generated from the Maelstrom rocket engine

Fluent Setup

1. Launch Fluent
    a. Close the Mesh window
    b. Go back to Ansys Workbench, and there should now be a lightning bolt next to Mesh
    c. Right-click Mesh and hit Update, wait until it turns into a green check mark

   d. Double-click Setup
   e. Set Solver Processes to 4
2. Models
   a. Under Setup, double-click Models, then double-click Energy, then check the Energy Equation box to turn it on
   b. Double-click Viscous and select the k-epsilon model
      i. Under k-epsilon, select “Realizable.”
      ii. Under Near-Wall Treatment, select “Enhanced Wall Treatment.”
3. Compressible Air Properties
   a. Under Setup, double-click Materials, then Fluid, and then air
   b. Change density to “ideal-gas.”
   c. Change Viscosity to “sutherland.”
   d. Hit “ok” and the “Change/Create.”
4. Boundary Constraints
   a. Under Setup, double-click Boundary Conditions and select “fuel_inlet.”
   b. Fuel inlet
      i. Change Type to “mass-flow-inlet” and then “edit.”
      ii. Set Mass Flow Rate to 0.02376 kg/s
      iii. Hit Apply
   c. Oxidizer inlets
      i. Select “ox_inlet”
      ii. Change Type to “mass-flow-inlet” and then “edit.”
      iii. Set Mass Flow Rate to 0.04752
      iv. Hit Apply
   d. Plenum Outlet
      i. Select “plenum_outlet”
      ii. Change Type to “pressure-outlet” and then “edit.”
      iii. Set Gauge Pressure to 0 and hit Apply
5. Set Up Goal Monitors
   a. Under Solution, right-click Report definitions > New > Surface Report > Area-Weighted Average
      i. Name it “max_chamber_pressure.”
      ii. Change field variable to “Total Pressure.”
      iii. Select “walls” and then hit “ok.”
   b. Right-click Report definitions > New > Surface Report > Mass-Weighted Average
      i. Name it “outlet_axial_velocity.”
      ii. Change Field Variable to Velocity… and Axial Velocity
      iii. Select “plenum_outlet” and then hit “ok.”
   c. Right-click Report definitions > New > Volume Report > Volume Average

        i. Name it “average_tke.”
        ii. Change the Field Variable to Turbulence… and Turbulent Kinetic Energy (k)
        iii. Select the volume and then hit “ok.”

6. Pseudo-Transient Setup
    a. Click Methods on the tree
    b. Make sure Scheme is coupled
    c. Ensure the Pseudo Time Method is on
7. Run and Iterate
    a. Click Initialize in the tree and then hit “Initialize.”
    b. Click Run Calculation and set the number of iterations to 300
    c. Click Calculate

Post Processing

1. Chamber Velocity Contours
    a. Results > Graphics > Contours > New
    b. Select Velocity… and Velocity Magnitude
2. Turbulent Kinetic Energy Profiles
    a. Results > Graphics > Contours > New
    b. Select Turbulence… and Turbulence Kinetic Energy (k)

Opening and Running Other Cases

1. Download the Maelstrom.wbpz file from our Canvas page
2. Open a blank instance of Ansys Workbench
3. Click File, Restore Archive…
4. Select Maelstrom.wbpz and unzip it
5. Double-click on Parameter Set
6. Look at the table called Table of Design Points. There are two columns, one for each diameter of the different holes. DP0 represents the baseline numbers.
7. For new cases, just click on the empty row below and input the new diameters.
8. Make sure to check the Retain box
9. You can input multiple cases at a time, and they will run in the background. You just need to update that table with all the cases you want, then hit Ctrl and select all the cases. After that, you right-click and hit Update Selected Properties
10. A green check mark will show up under Retained Data when it is finished; this could take a while depending on how many simulations you’re running, but it all happens in the background.

11. Right-click on your new case (DP 1, for example) and select Set as Current. Go back to the main project tab and double-click the Results cell. From here, you will be able to see all of the outputs. Refer to the Post Processing section to get all the contours. Once you have all the data, you can close that window and repeat these steps for however many cases you did.

Results

Control

- Fuel Inlet: 3.5131 mm
- Oxidizer Inlet: 2.19 mm

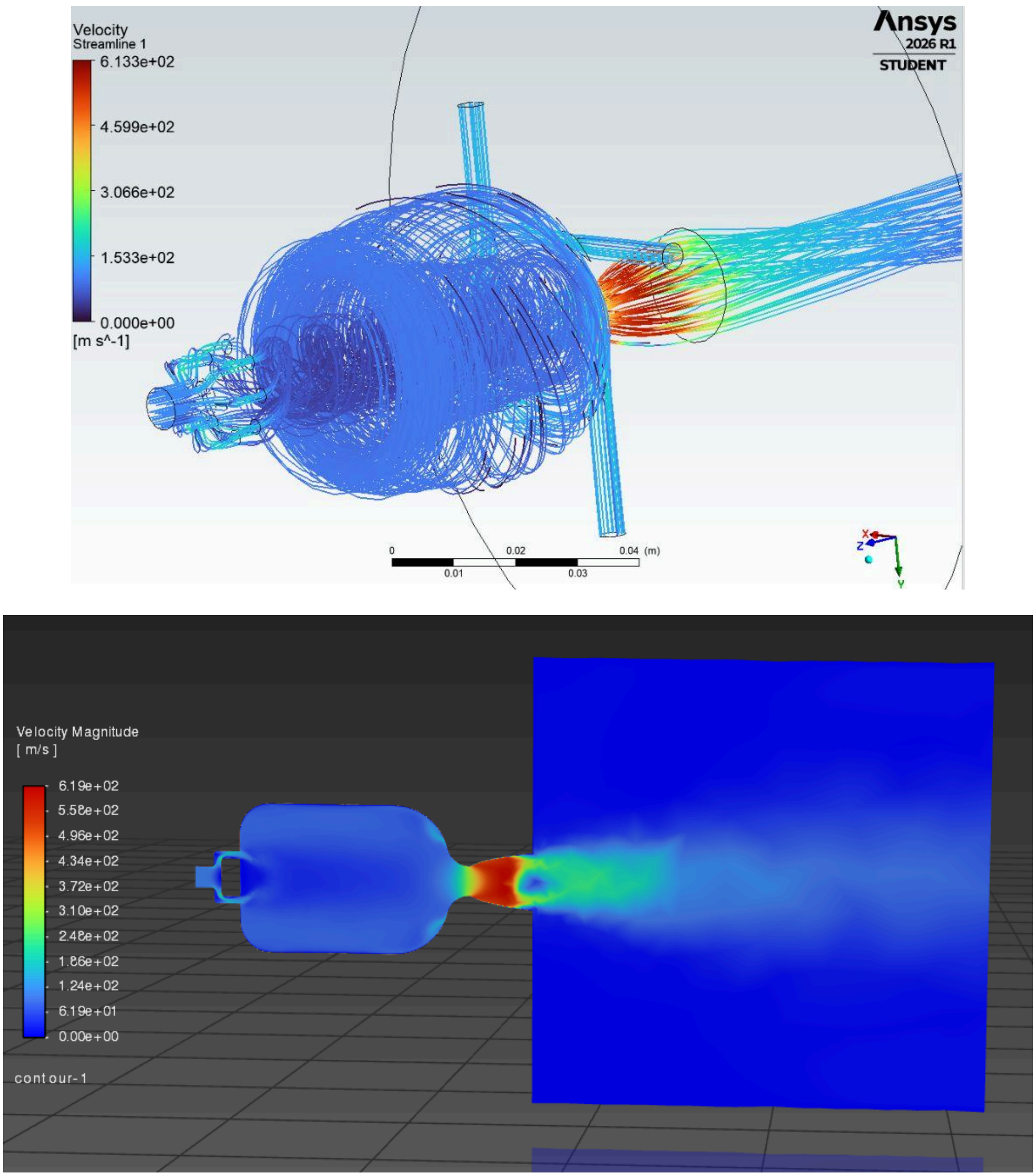


Figure 3: Top: velocity streamlines, Bottom: center plane contours

The first image of Figure 3 displays the velocity streamlines generated by the fuel and oxidizer inlets, consisting of the original fuel inlet of 3.51 mm and oxidizer inlet of 2.19 mm radii. It progresses forward through the combustion chamber, where the streamlines indicate that the oxidizer injections successfully generate rotational motion as it follows a spiral path, further indicating vortex formation. The flow, initially around 153.7m/s in the chamber, then increases to hypersonic velocities at a max of 614.7m/s as it converges at the nozzle, which is physically consistent with the conservation of mass principle, $\dot{m} = \rho AV$ where flow acceleration near the throat is expected for converging nozzle geometries. This occurs because mass flow rate must be conserved as the fluid accelerates as it approaches the throat, thus producing the high-velocity region best seen in the second image, which displays the velocity contours in Figure 3. This control simulation of the default values provides the following scenarios with a grounded check of what should be expected and can be compared to see what deviates.

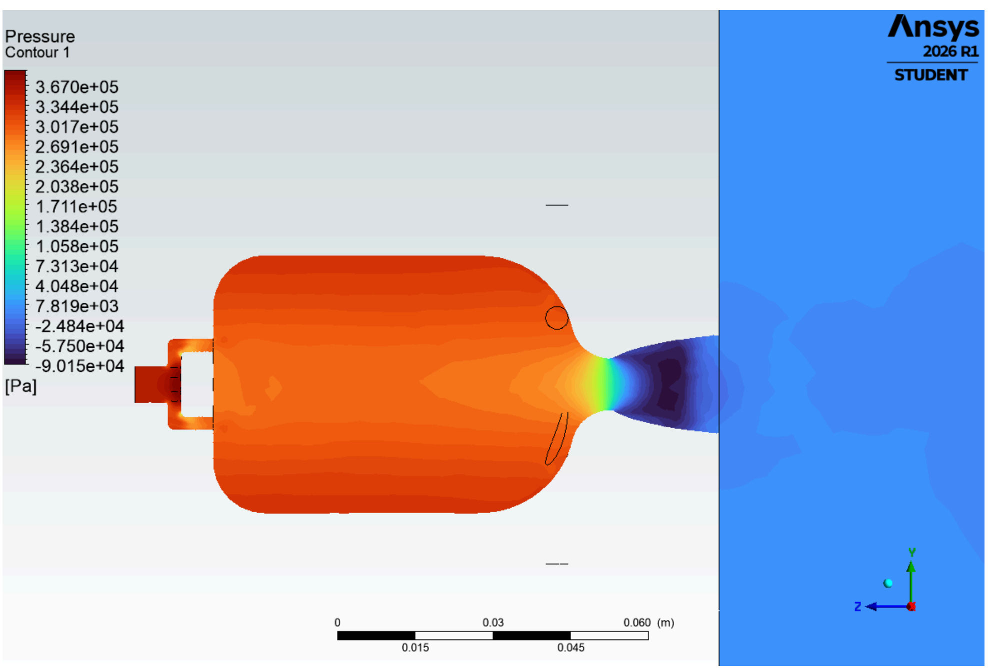


Figure 4: Center plane pressure contour

Figure 4: The pressure contour here displays that the highest pressure occurs inside of the combustion chamber around 301kPa and decreases significantly to around 105 kPa at the throat. This trend makes physical sense as the expected pressure energy is converted into kinetic energy as the flow accelerates through the converging section of the nozzle. The pressure then leaves the

plenum outlet, which corresponds to the applied 0 Pa gauge. This means that the blue region exiting the plenum indicates the atmospheric outlet conditions.

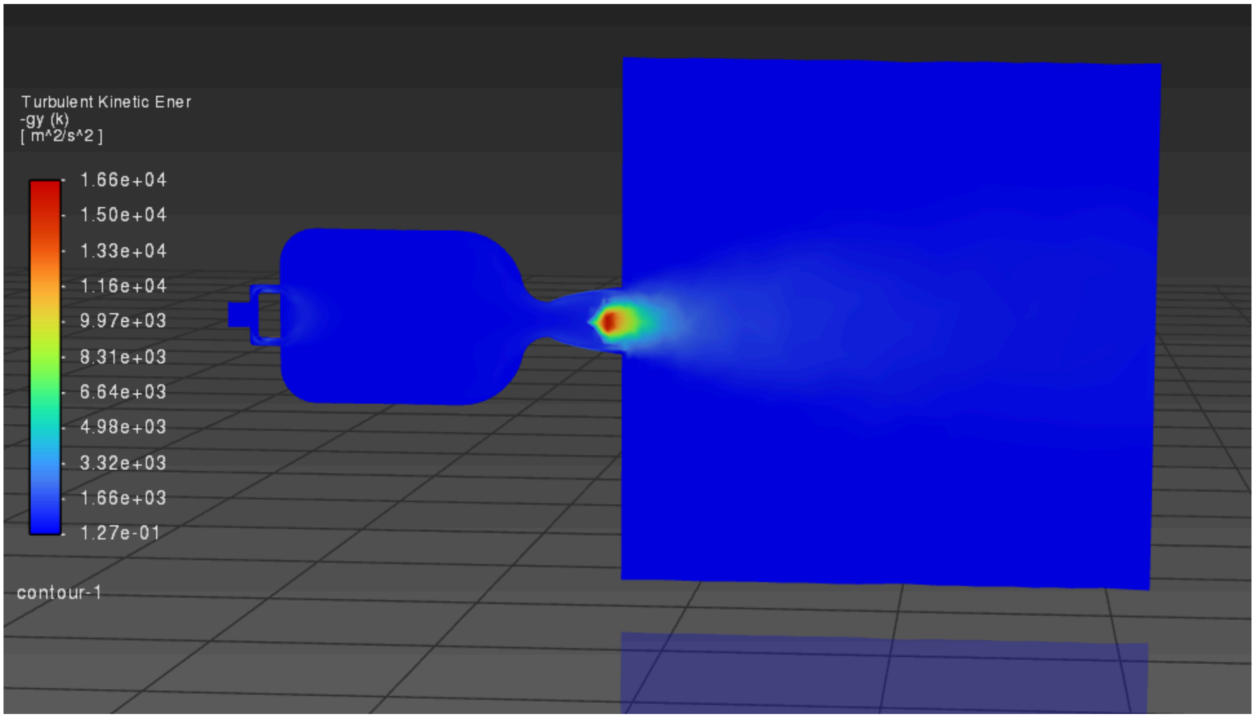


Figure 5: Center plane turbulence kinetic energy contour

The TKE contour reaches a maximum of 17,000m^2/s^2, with the highest turbulence concentrated near the nozzle throat. Most of the chamber remains near the lower end of the scale, which shows that the strongest turbulence production occurs when the flow is accelerated and experiences the largest velocity gradient in the nozzle. This displays an agreement with the previous velocity contours, where the highest speeds are also located near the throat. This directly relates to the equation, $k = \frac{1}{2}(u'^2 + v'^2 + w'^2)$, where turbulent kinetic energy represents the energy content within velocity fluctuations. This means that regions of high TKE occur near the nozzle entrance where the oxidizer and fuel interact, merging the two velocity regions where shear layers are the strongest.

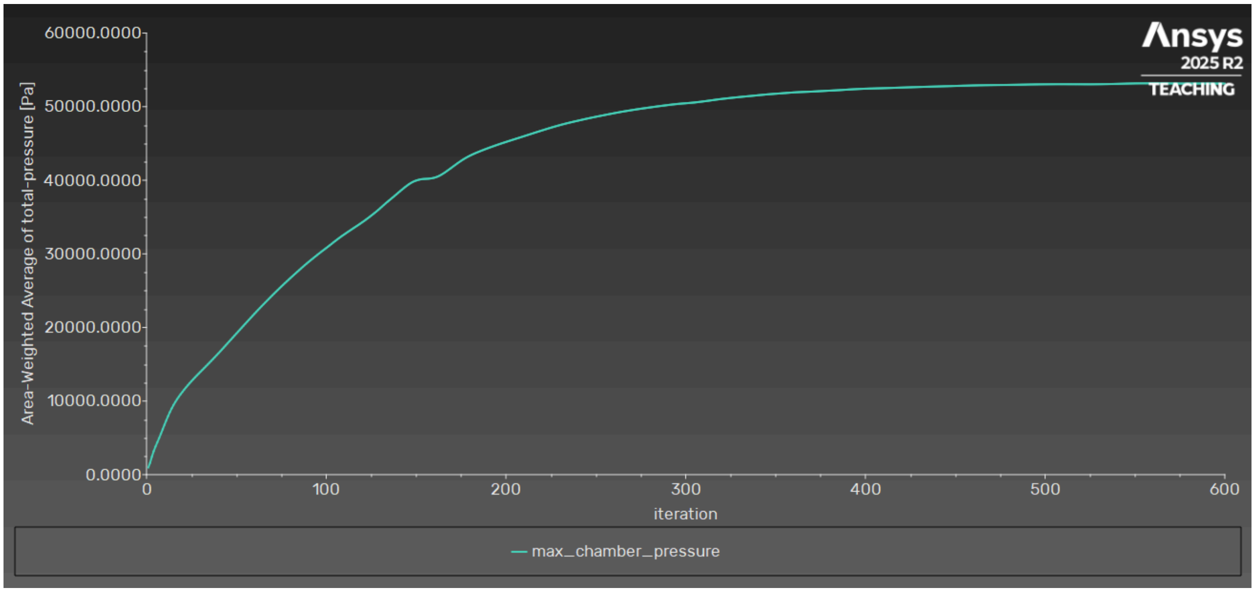


Figure 6: Max chamber pressure vs iteration

The chamber pressure vs iteration figure displays a convergence to about 53 kPa after 400 iterations. The asymptotic behavior of the curve indicates that the solution is reaching a steady-state condition. The large absence of oscillation at the final iteration also suggests that the pressure has stabilized and the solution has successfully converged

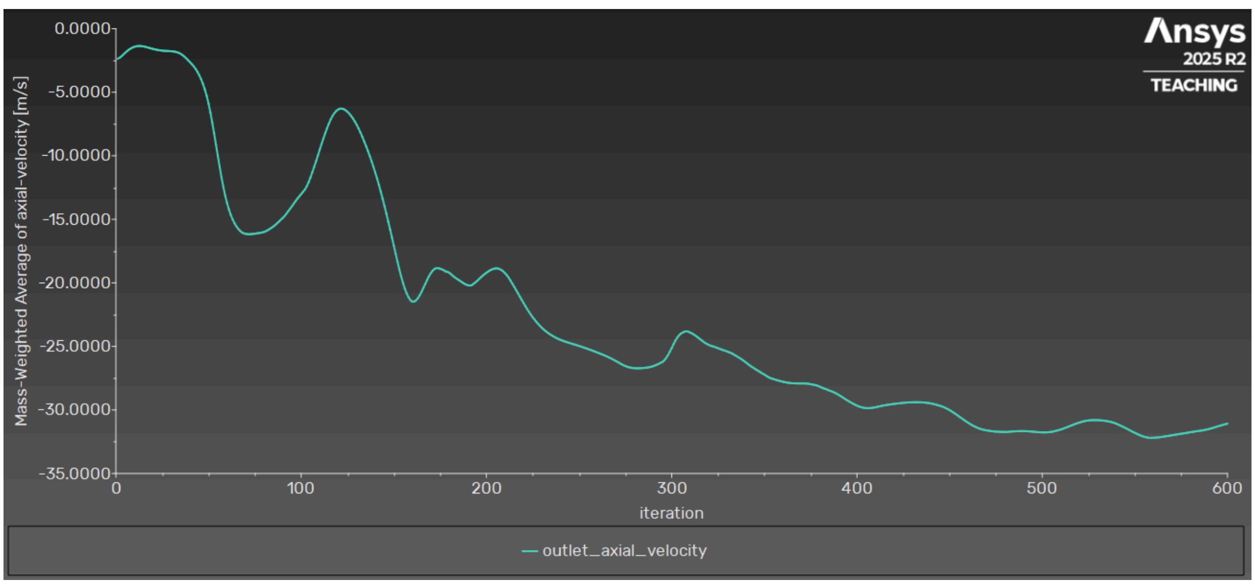


Figure 7: Mass-weighted average outlet axial velocity vs iteration

This figure shows that the average axial velocity converges to a nearly constant value of 31m/s (which displays as negative, likely due to our coordinate system) as the number of iterations increases. This behavior displays that the solver reached a steady-state solution and the outlet flow characteristics are no longer changing significantly, displaying successful convergence for post-processing and analysis.

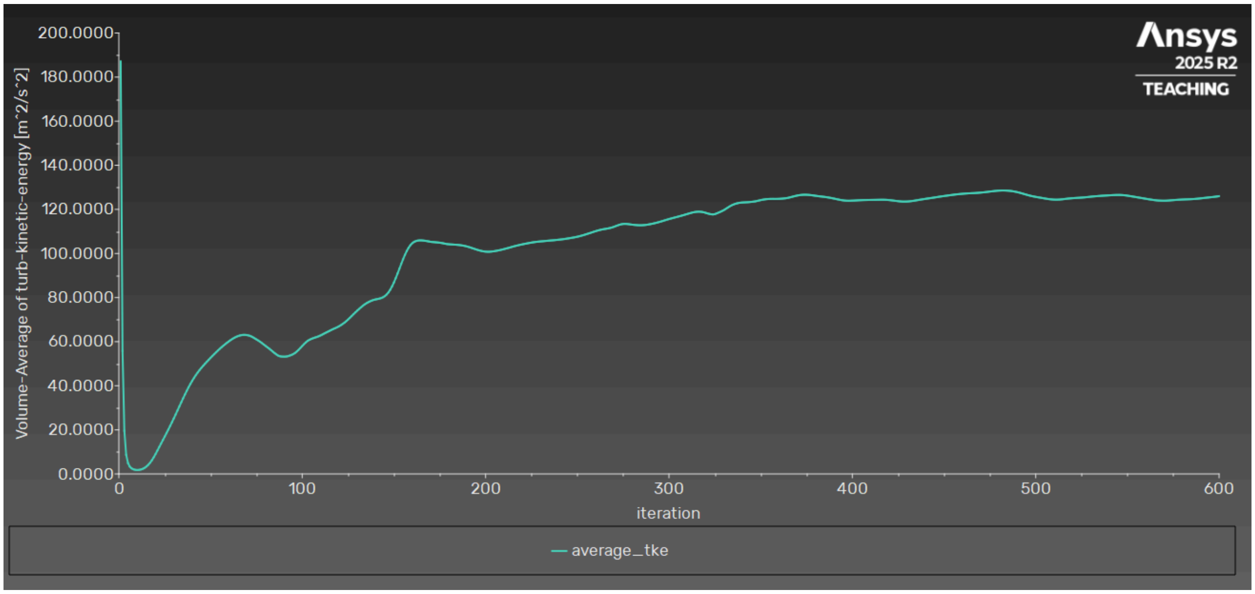


Figure 8: Average TKE

Figure 8 shows the trendline of the average TKE with respect to the iteration count. The increase in turbulence is a direct reflection of the vortex motion and shear-driven mixing within the combustion chamber. Once again, after 350 iterations, we can see that the value stabilizes, which indicates full development and a steady-state solution.

Scenario 1, Oversized Configuration:

- Fuel Inlet: 3.86 mm
- Oxidizer Inlet: 2.41 mm

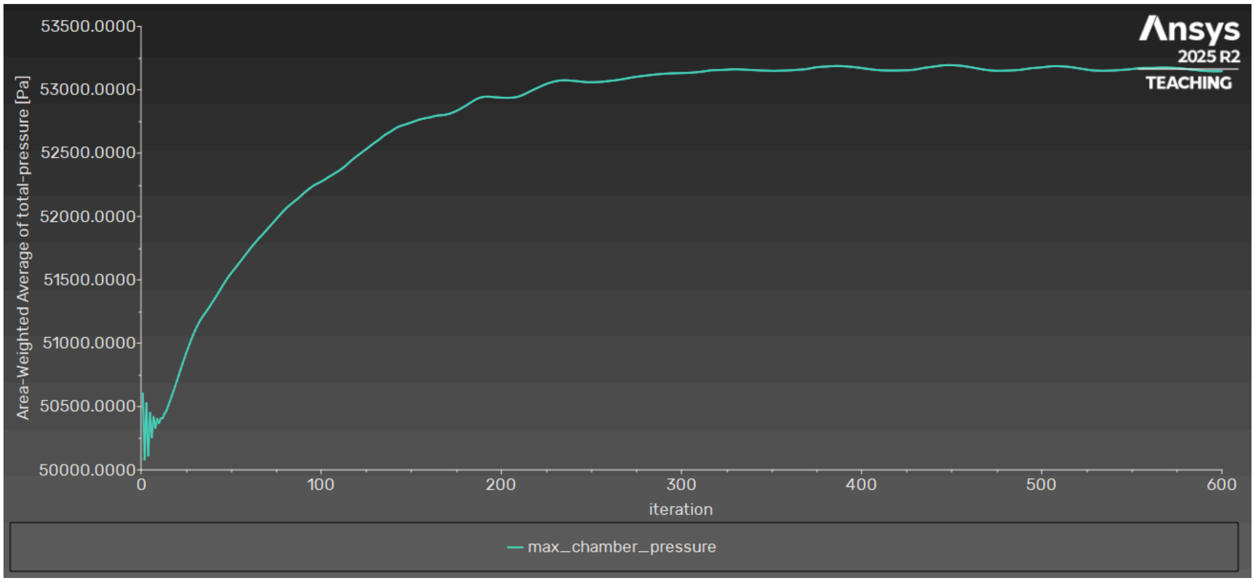


Figure 9: Max chamber pressure vs iteration

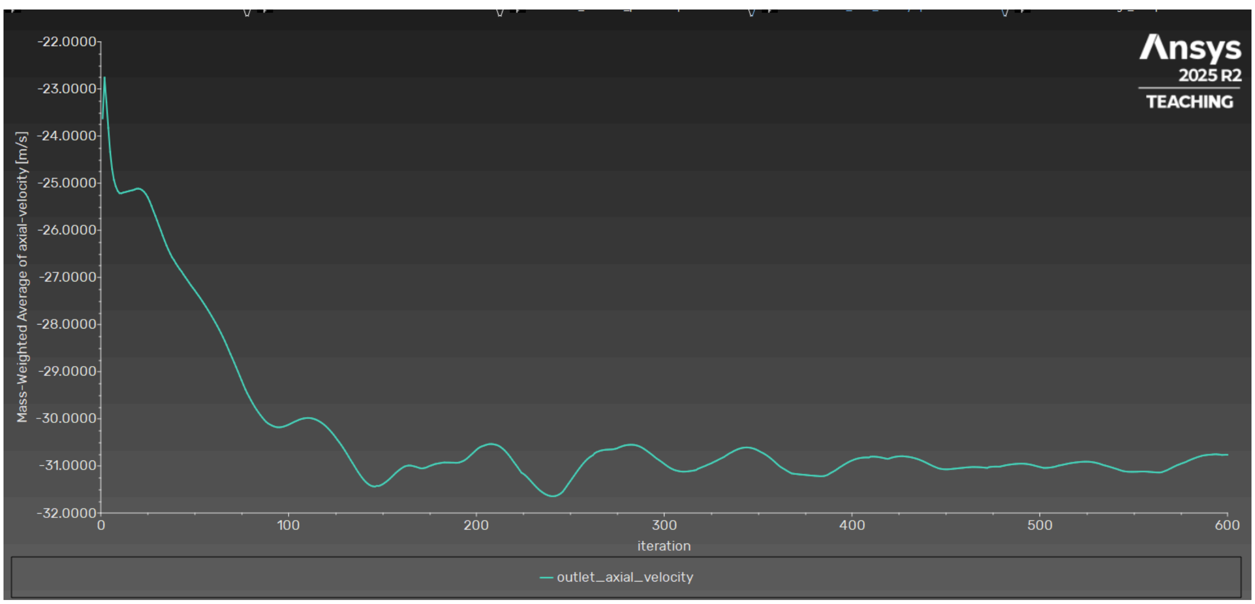


Figure 10: Mass-weighted average outlet axial velocity vs iteration

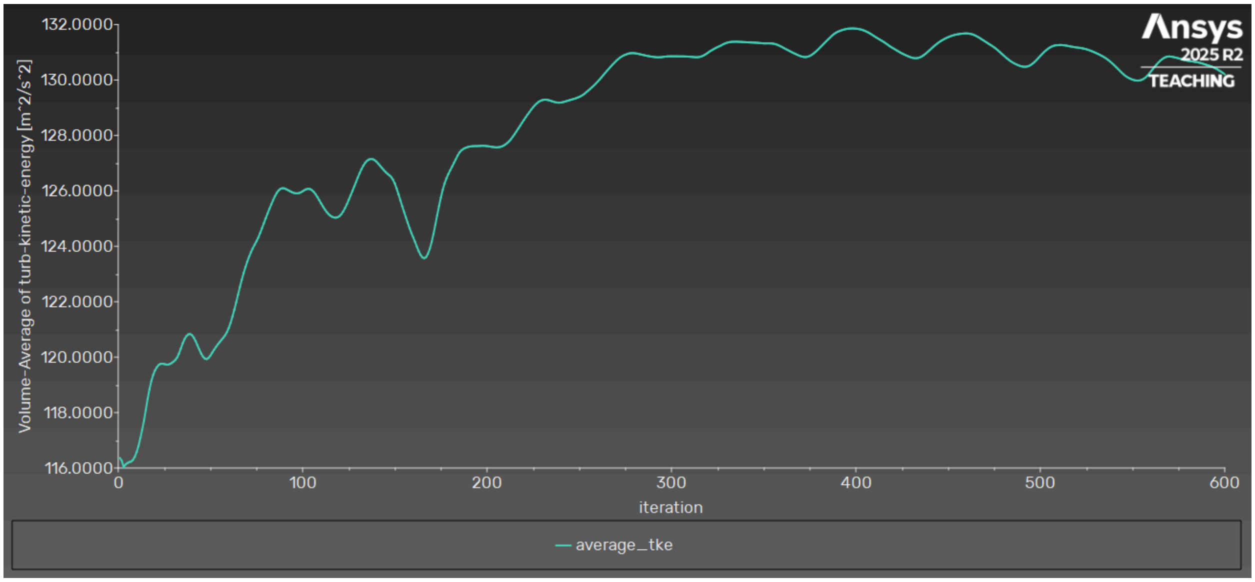


Figure 11: Average TKE

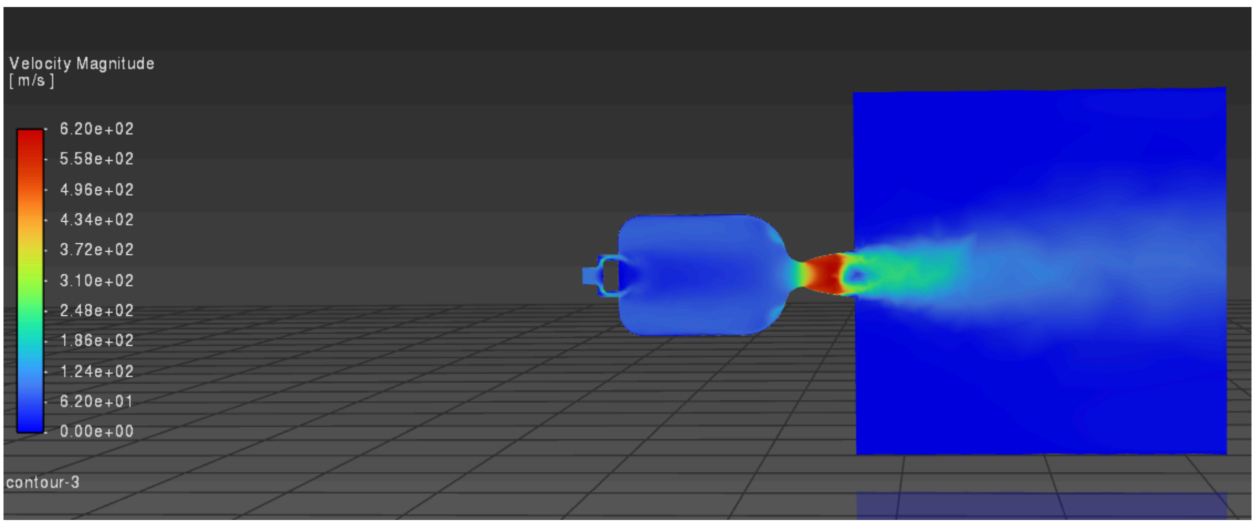


Figure 12: velocity magnitude contour

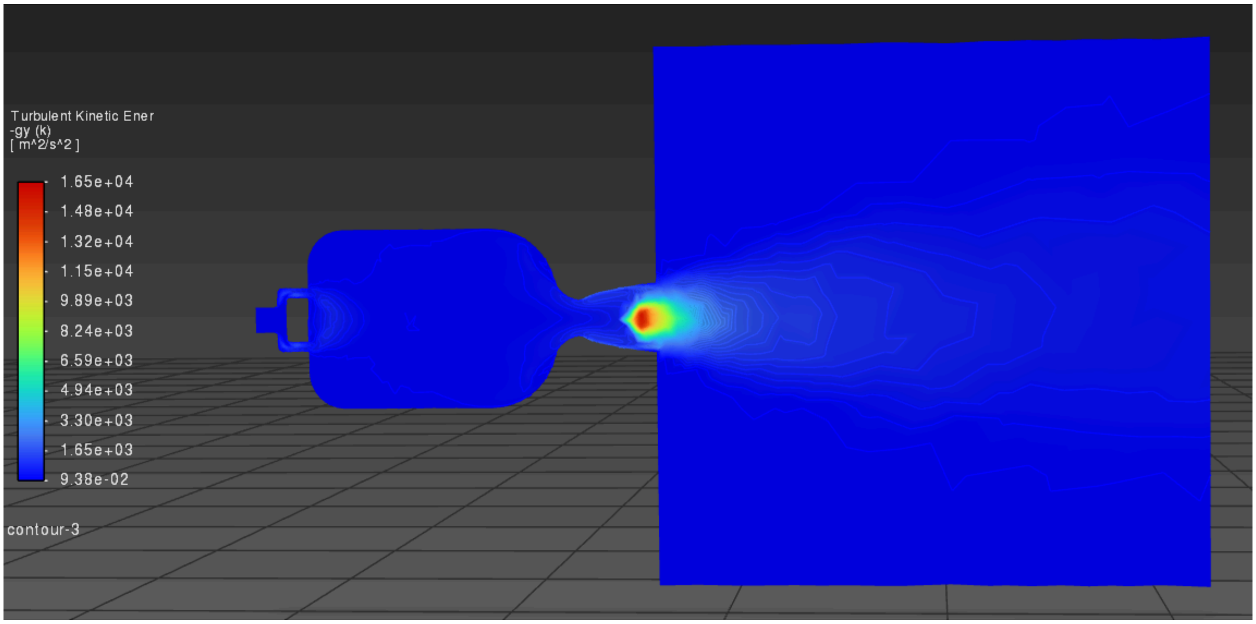


Figure 13: TKE contour

A common theme we will see in the following scenarios is that the contour plots provide very little information on the differences between each scenario due to the scenarios having very slight variances in the inputs. The iteration plots tell a better story.

The small differences between the oversized and control configurations are also physically reasonable because a 10% increase in the inlet's radius only produces a moderate change in the area for flow to pass through. Since the inlet area varies, $A = \pi r^2$, a 10% increase in radius will increase the area by 21%. For a fixed mass flow rate, we note that the continuity equation, $\dot{m} = \rho A V$, tells us that increasing the flow area will also reduce the inlet velocity. This will result in the oxidizer jets entering the chamber with less momentum, which reduces the overall strength of the vortex.

Another notable outcome of the oversized holes scenario is that the maximum chamber pressure and average TKE tend to have greater fluctuations over the iterations, while the axial velocity seems to converge more quickly. With that being said, almost all values converged to near-equivalent values, where the oversized configuration produced a maximum chamber pressure of **~53 kPa,** an outlet axial velocity of **32 m/s,** and an average TKE of 130 m^2/s^2 compared to 124 m^2/s^2 average TKE from the control. This slight increase in average TKE occurs because of the modification done on the injector's geometry, which helped to promote turbulent mixing. However, the pressure and velocity at the outlet remain nearly unchanged as the flow behavior is similar to the control's configuration. The visual difference in the average TKE can be attributed to a weaker vortex, which means more premature mixing of the fuel and oxidizer prior to reaching the injector side of the chamber. Overall, the oversized hole scenario suggests that a 10% increase in the radius does not significantly alter the vortex structure or engine performance.

Scenario 2, Undersized Configuration:

- Fuel Inlet: 3.16 mm
- Oxidizer Inlet: 1.97 mm

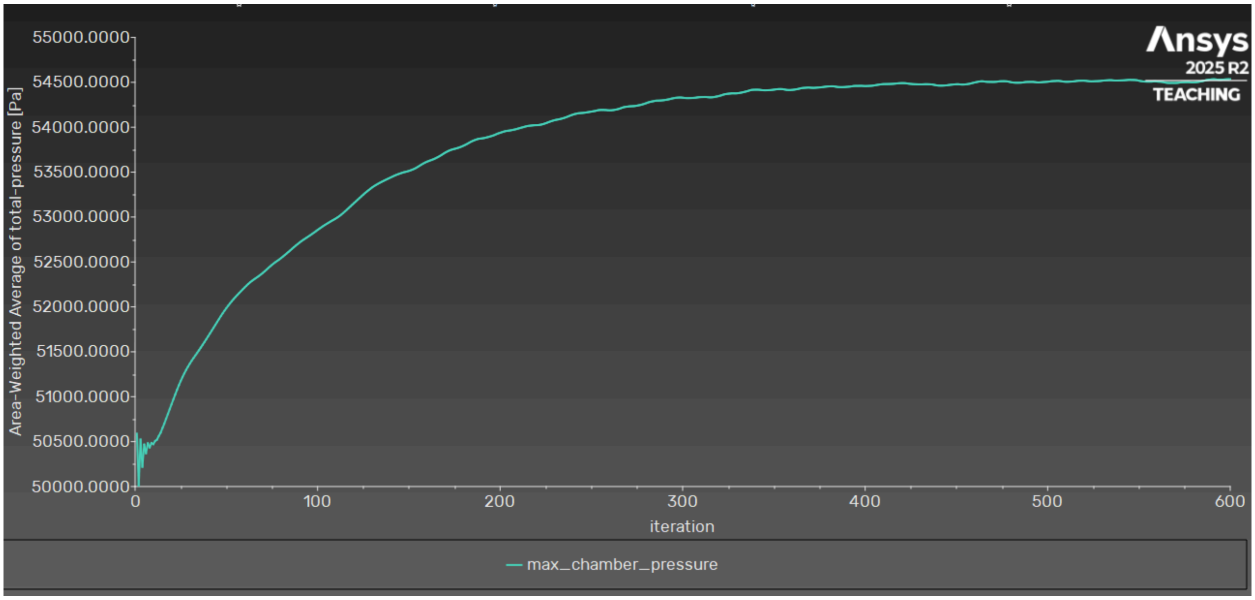


Figure 14: Max chamber pressure vs iteration

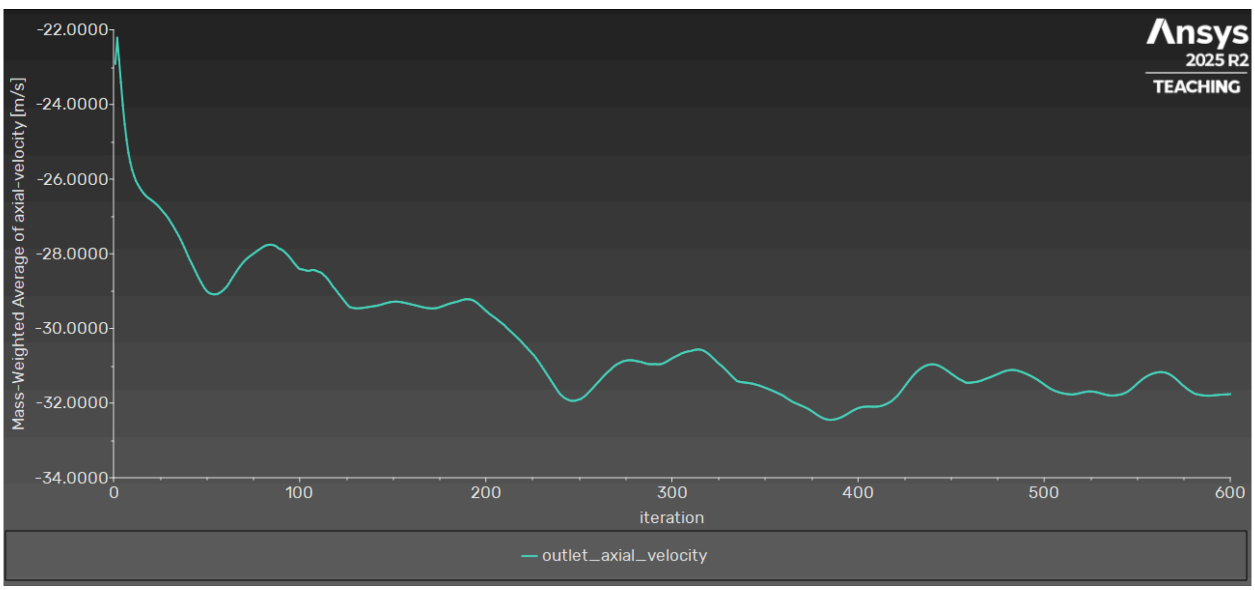


Figure 15: Mass weighted average outlet axial velocity vs iteration

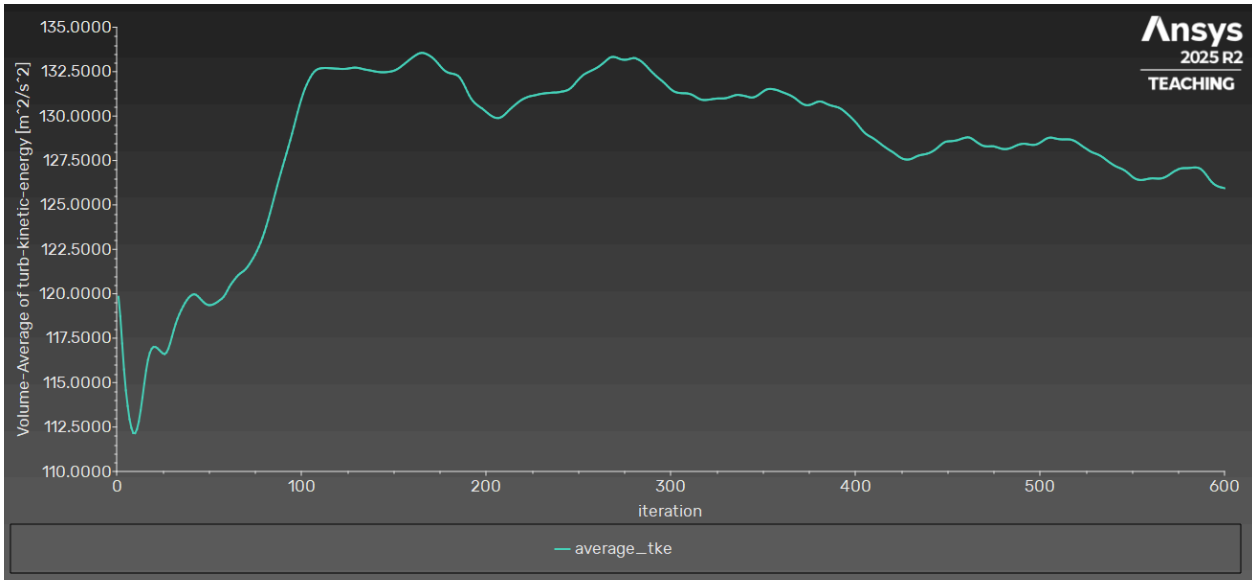


Figure 16: Average TKE

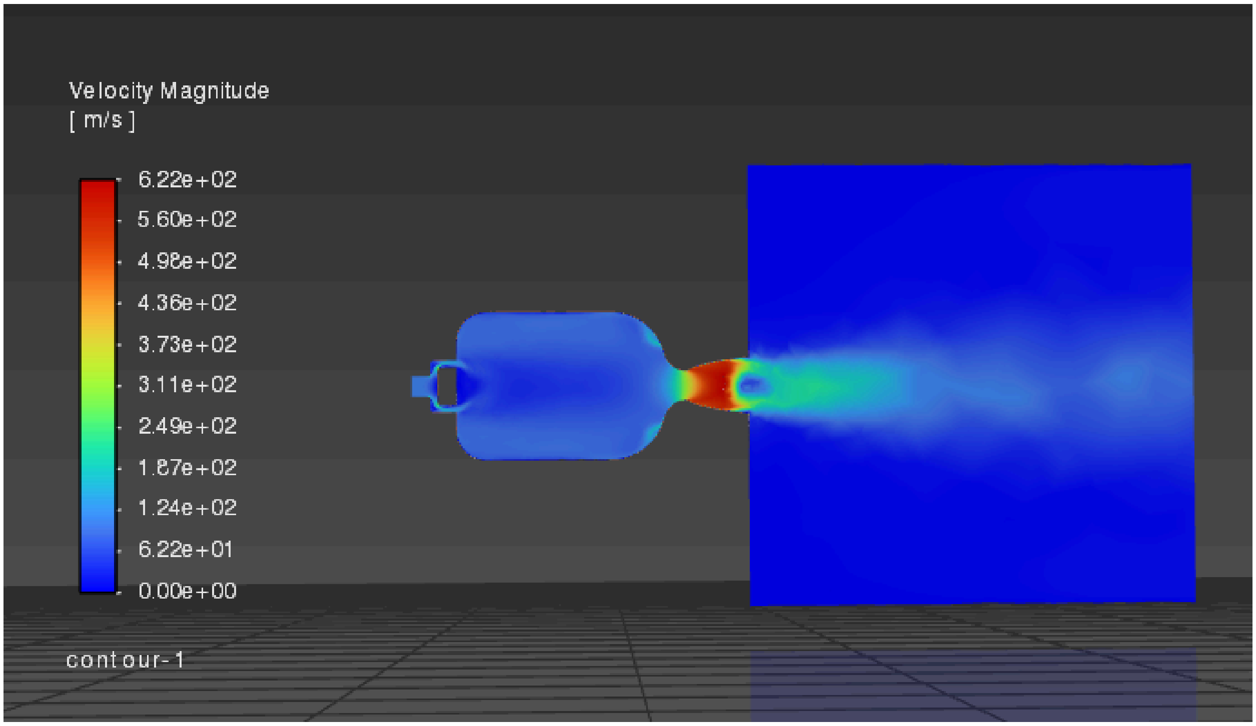


Figure 17: velocity magnitude contour

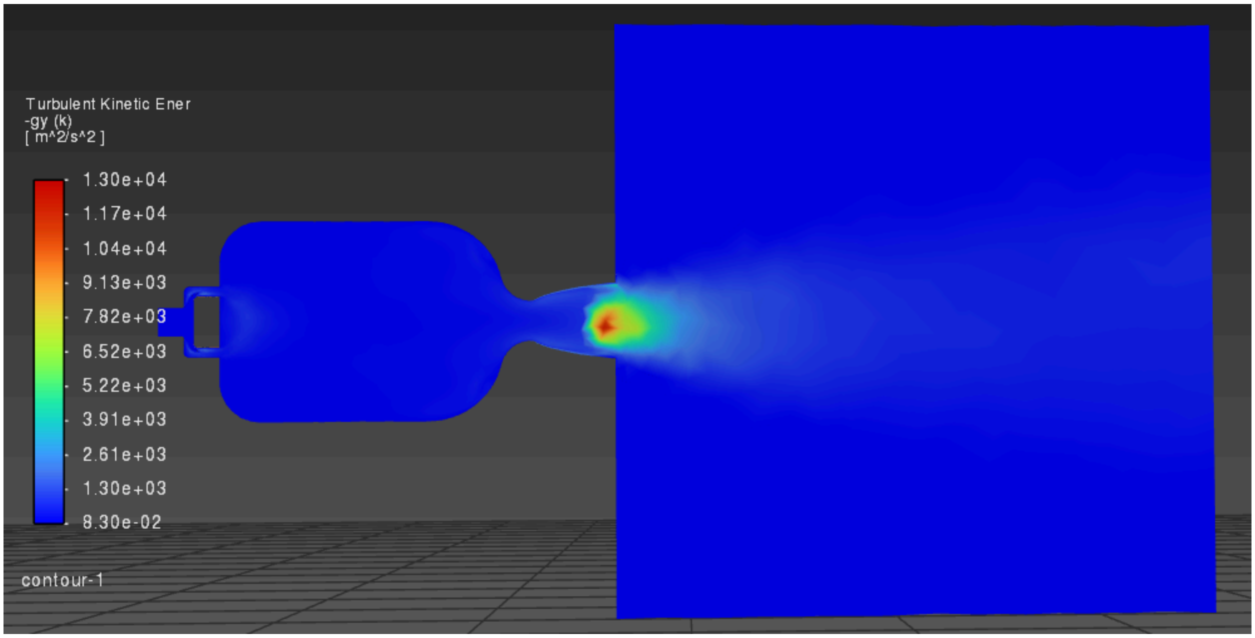


Figure 18: TKE contour

Control: ~53 kPa max chamber pressure, 32 m/s outlet axial velocity, 124 m^2/s^2 average TKE.

The outlet velocity and maximum chamber pressure appeared visually identical to those of the control configuration. The pressure remained around 53kPa, which suggests that the reduction of the radius primarily affects the flow acceleration rather than the pressure itself. The average TKE was shown to be around 126 m^2/s^2 at iteration 600, but followed a steady decreasing trend rather than leveling off. This indicates that the turbulence field did not converge even though pressure and velocity solutions had stabilized. As a result, the reported TKE value should be taken as an approximation of the converged solution.

The increase in TKE indicates a greater amount of turbulent mixing between the fuel and oxidizer. This occurred because the higher velocity observed in the undersized configuration is physically reasonable, as reducing the injector radius decreases the available flow area. We can see that the maximum velocity in the control is 619m/s in comparison to the 622m/s of the undersized inlet radius. According to the continuity equation, $\dot{m} = \rho A V$, decreasing the area will increase the velocity, which leads to the oxidizer jets entering the chamber with greater momentum, strengthening the vortex motion.

Overall, the undersized hole scenario produces slightly stronger mixing and higher velocity while preserving the general vortex structure, pressure, and flow behavior observed in the control configuration. If the inlet radii are further reduced, this design approach could be utilized to improve mixing between the fuel and oxidizer streams and to strengthen local vortex motion without significantly changing other physical properties.

Scenario 3, High Shear:

- Fuel Inlet: 4.04 mm
- Oxidizer Inlet: 1.86 mm

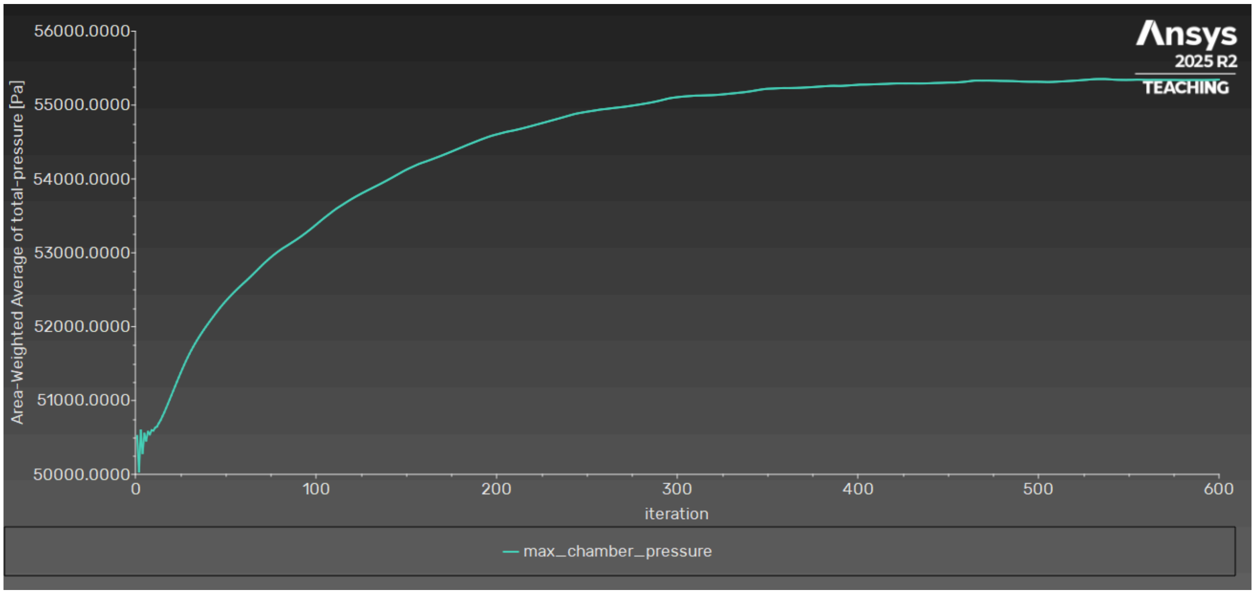


Figure 19: Max chamber pressure vs iteration

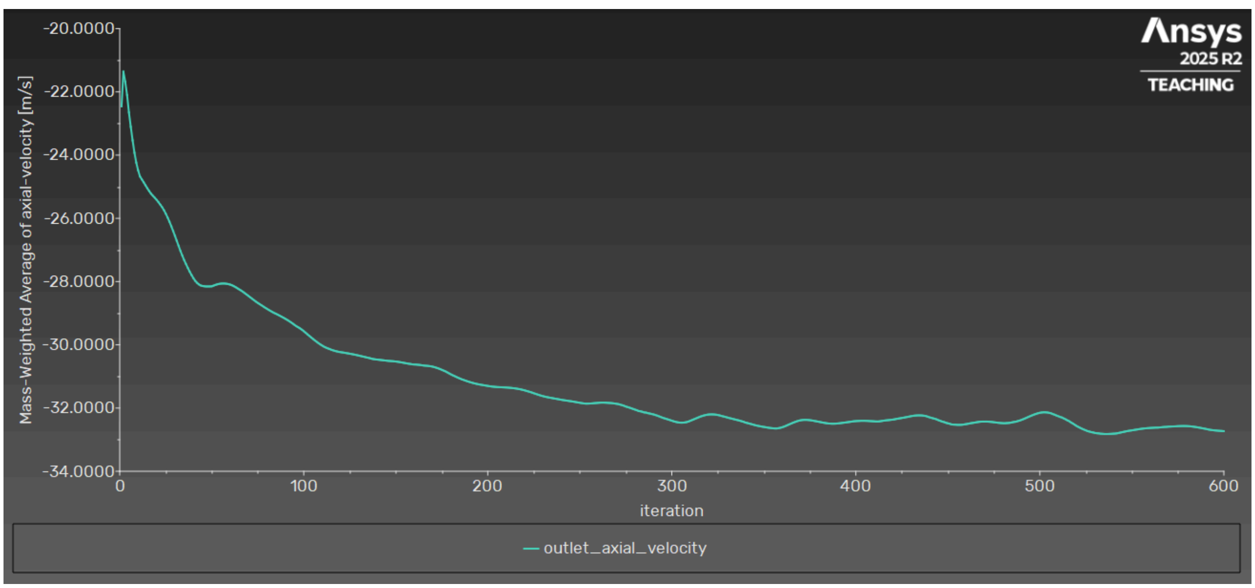


Figure 20: Mass weighted average outlet axial velocity vs iteration

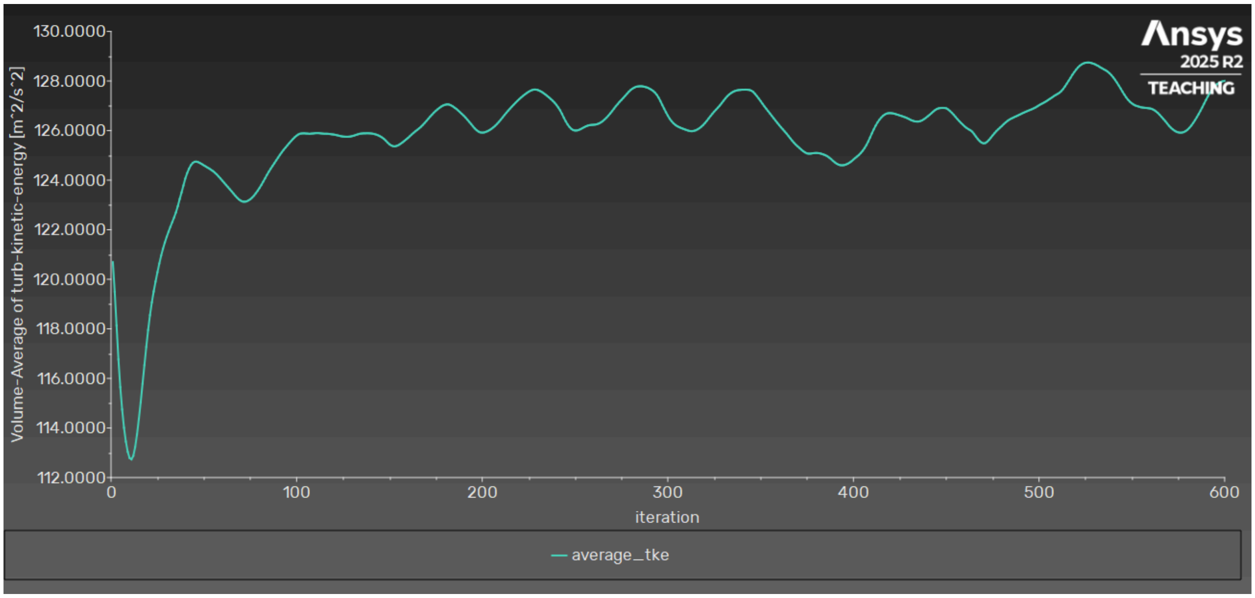


Figure 21: Average TKE

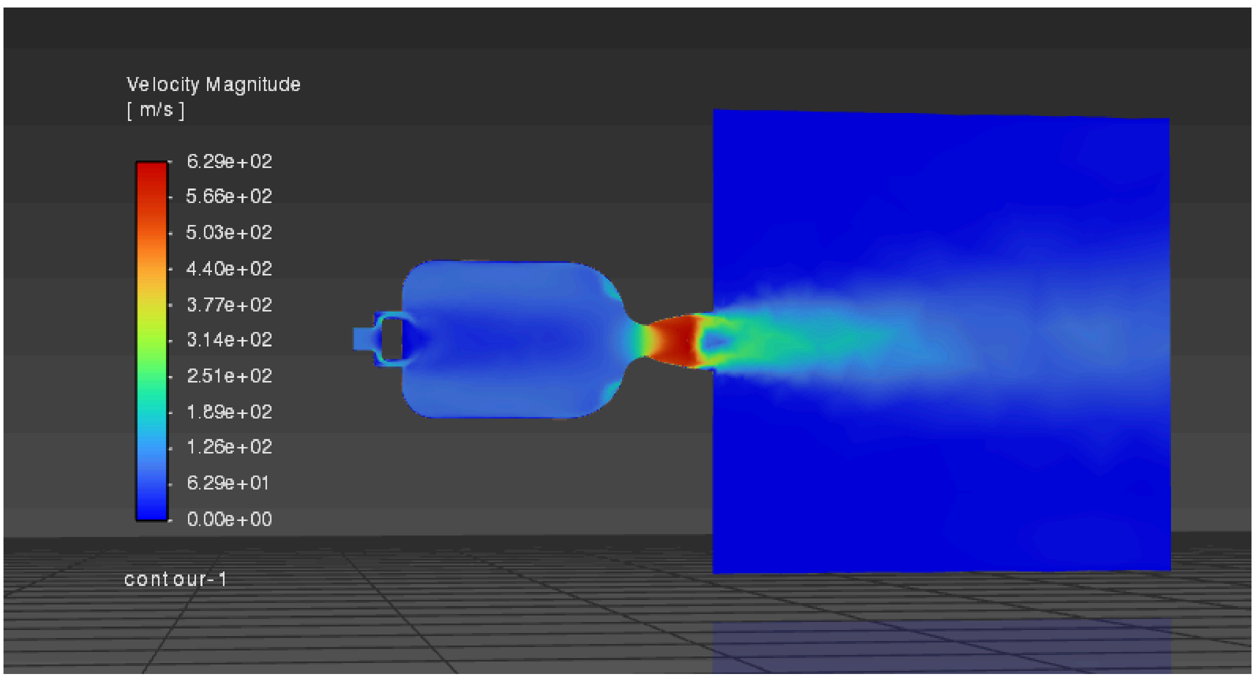


Figure 22: velocity magnitude contour

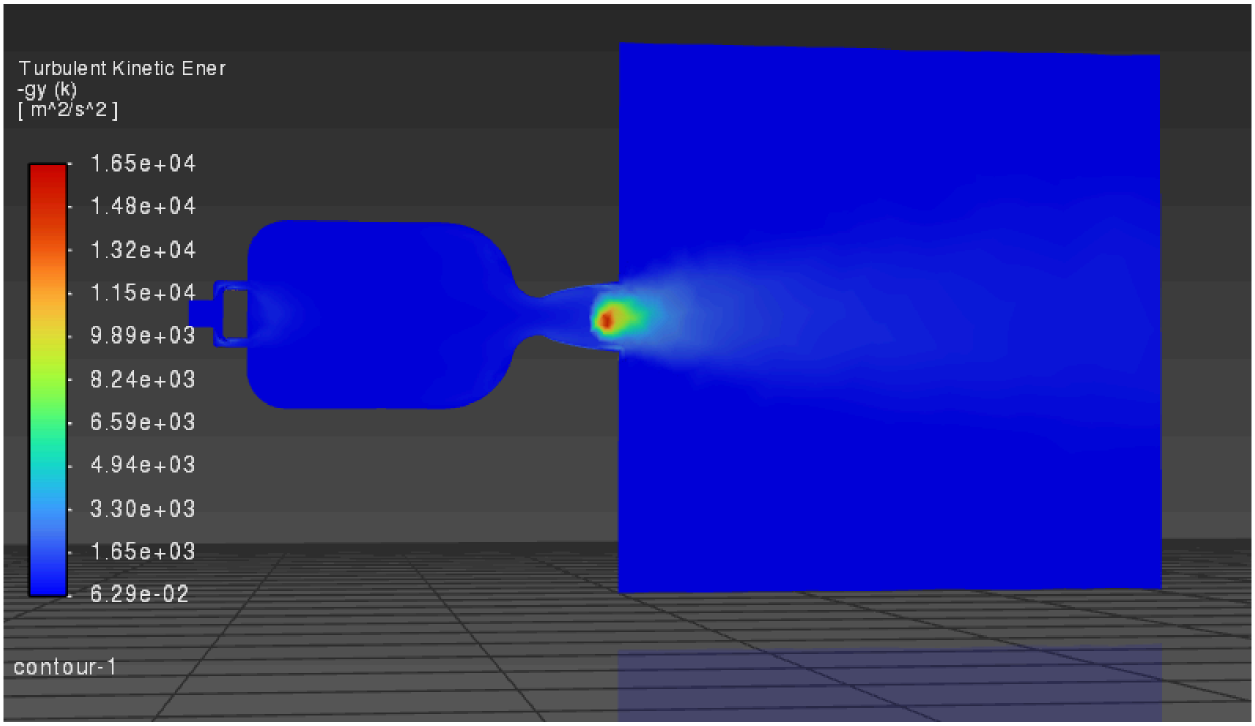


Figure 23: TKE contour

Control: ~53 kPa max chamber pressure, 32 m/s outlet axial velocity, 124 m^2/s^2 average TKE.

The high shear configuration was designed to increase the velocity difference between the oxidizer vortex and the central fuel jet by decreasing the oxidizer inlet radius while increasing the fuel inlet radius. This modification promotes a stronger shear layer at the interface between the two flow regions and was expected to increase turbulence generation as well as mixing within the chamber.

The max chamber pressure is ~55 kPa, which is higher than the control. This can be attributed to the fact that when the vortex is at a greater velocity differential to the fuel jet, there is increased shear, which increases the effective eddy viscosity, which can, in turn, require a greater chamber pressure in order to push the same amount of mass through the chamber. The outlet velocity and average TKE have a very marginal increase and more variability between iterations, possibly due to the added chamber pressure and more viscous interactions between the vortex and jet.

Overall, the high shear scenario increases the chamber pressure and turbulence generation while maintaining similar outlet flow as the control. This suggests that increasing the velocity difference between the vortex and fuel jet can support mixing without significant alterations to the overall flow structure.

Scenario 4, Core Punch:

- Fuel Inlet: 2.99 mm
- Oxidizer Inlet: 2.52 mm

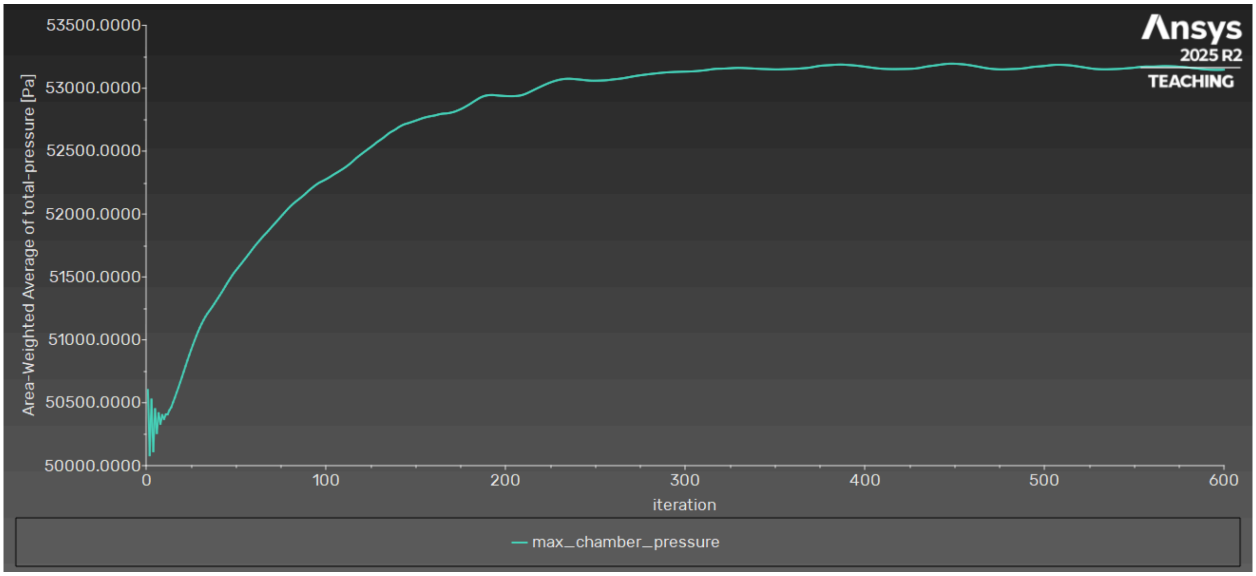


Figure 24: Max chamber pressure vs iteration

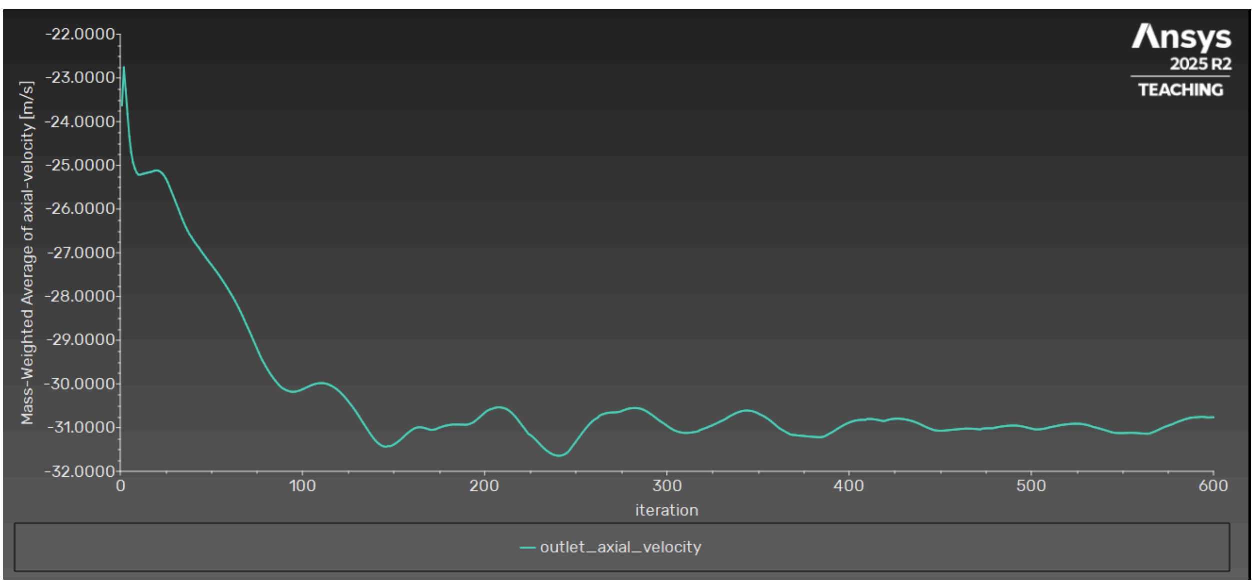


Figure 25: Mass-weighted average outlet axial velocity vs iteration

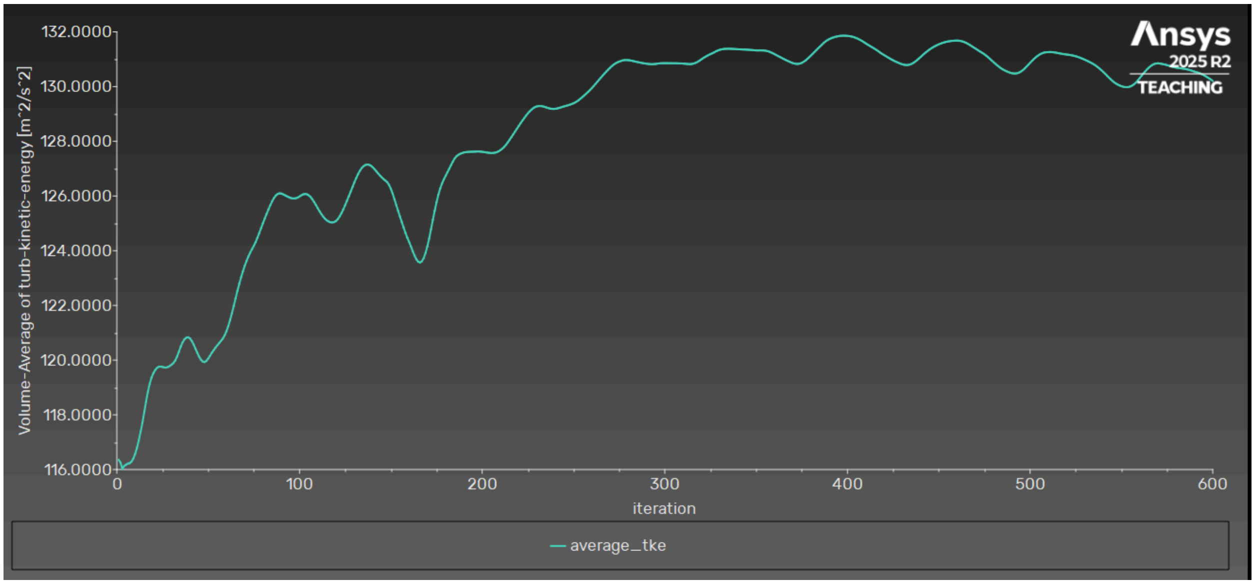


Figure 26: Average TKE

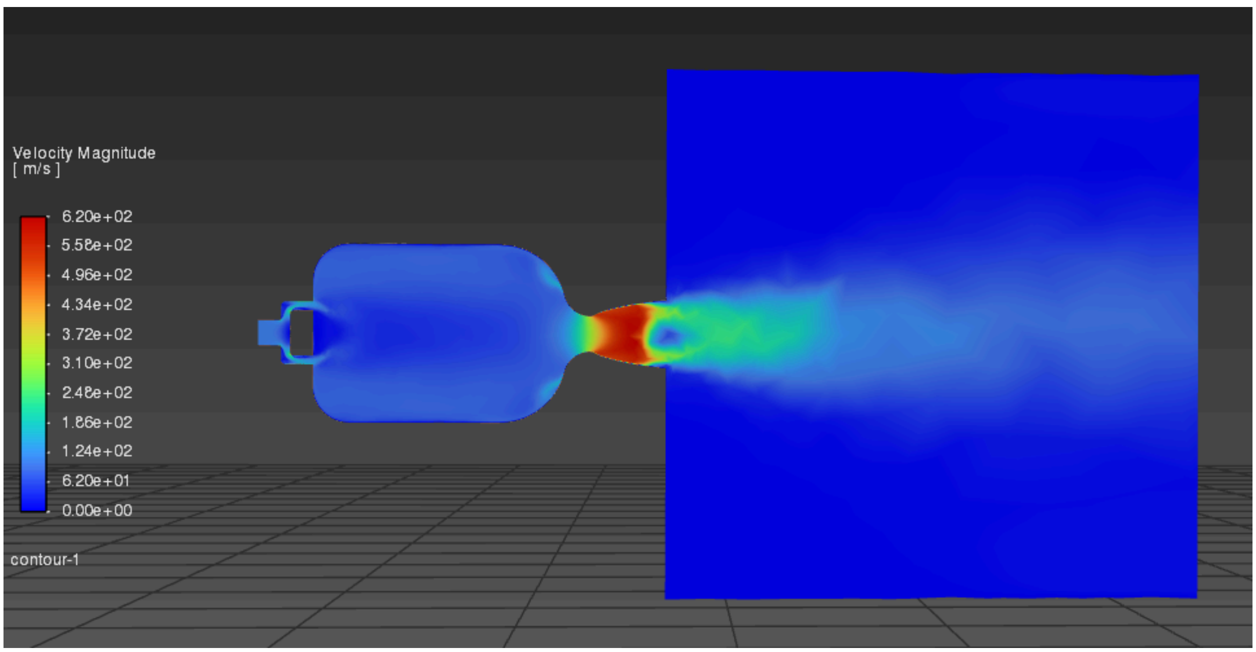


Figure 27: velocity magnitude contour

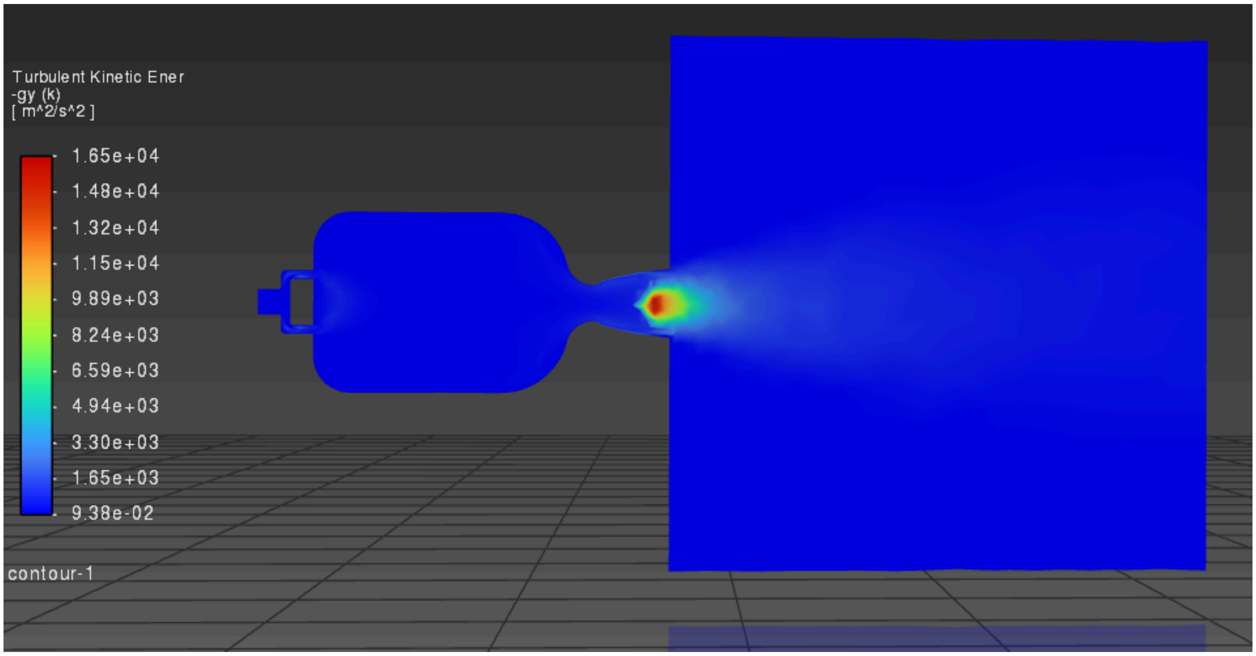


Figure 28: TKE contour

Control: ~53 kPa max chamber pressure, 32 m/s outlet axial velocity, 124 m^2/s^2 average TKE.

The core punch configuration was designed to accelerate the center fuel jet relative to the vortex by decreasing the fuel inlet change by 15% and increasing the oxidizer inlet by 15%. By creating this velocity mismatch, the resulting velocity should cause the fuel jet to penetrate much further into the center of the chamber, "punching" through the vortex.

The results of the average TKE converged to 130 $m^2/s^{2,}$ which is a considerable amount greater than the control. This can be attributed to the velocity differential between the jet and the vortex, similar to the high-shear scenario. Learning from this and the high shear simulations, we can conclude that a vastly different velocity mismatch between the jet and vortex will result in higher turbulence and thus affect the performance of the vortex cooling negatively, as well as decrease the efficiency of the engine.

Scenario 5, Velocity Match:

- Fuel Inlet: 3.51 mm
- Oxidizer Inlet: 2.48 mm

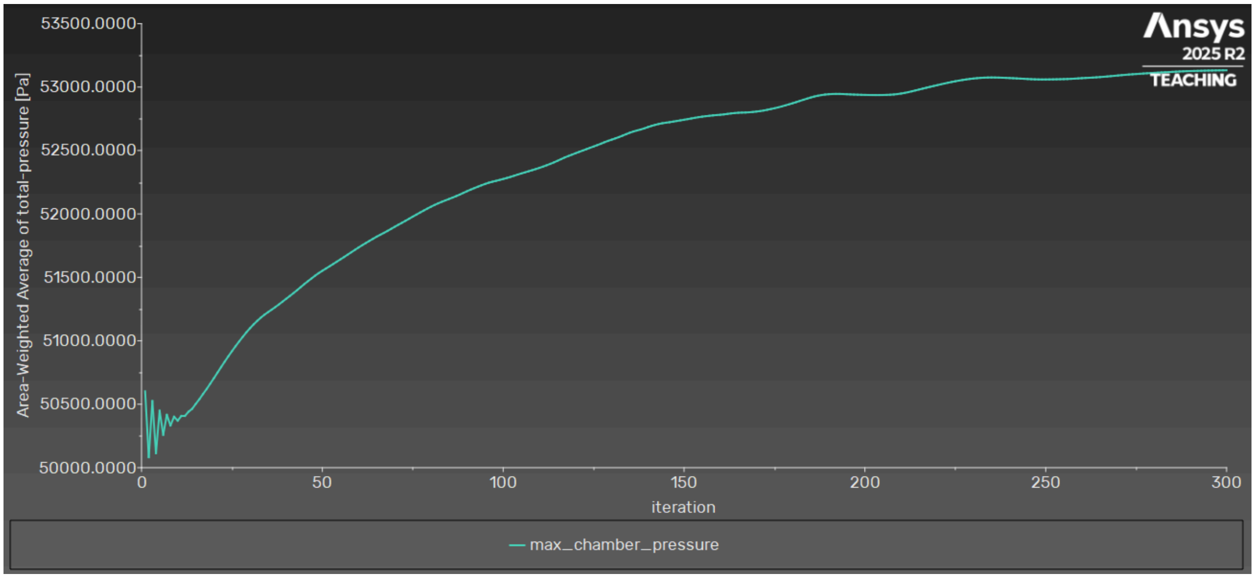


Figure 29: Max chamber pressure vs iteration

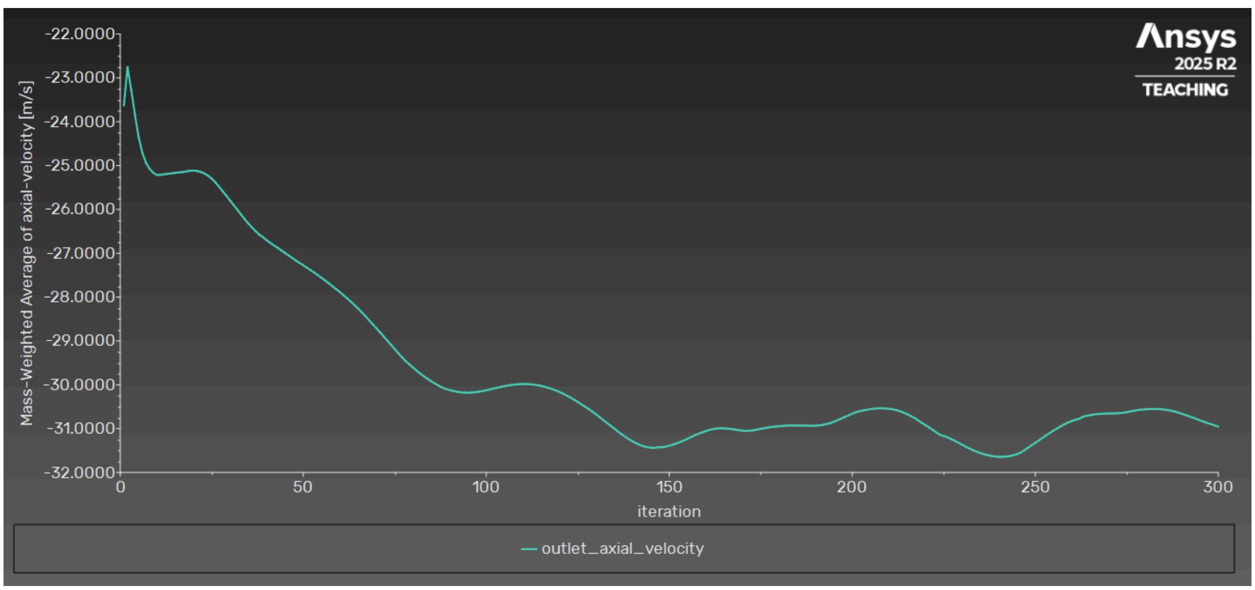


Figure 30: Mass weighted average outlet axial velocity vs iteration

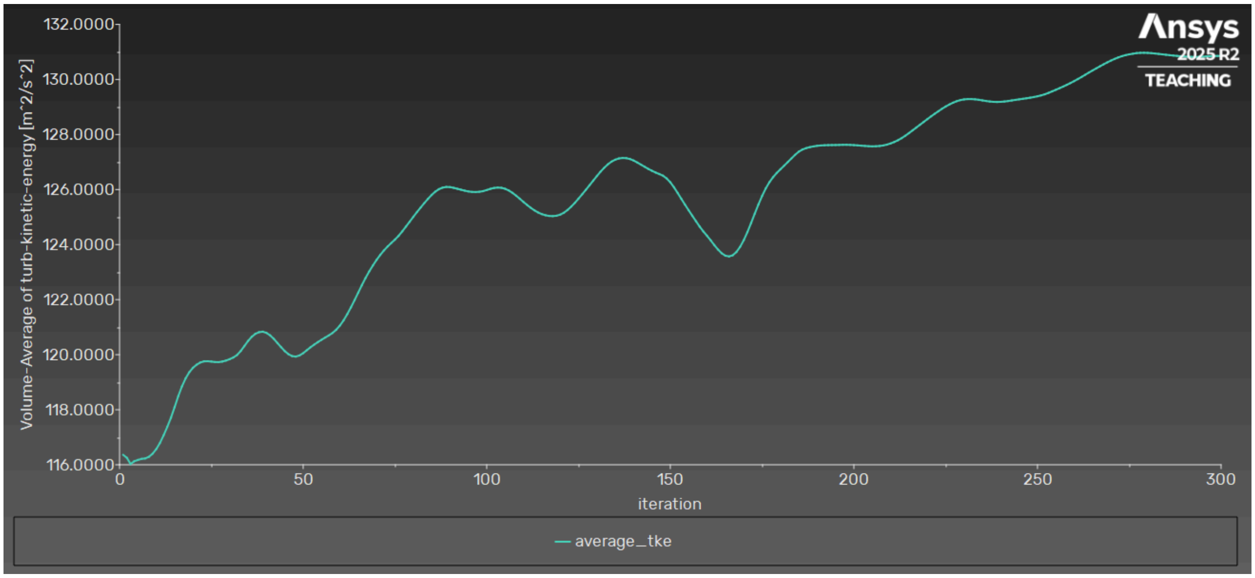


Figure 31: Average TKE

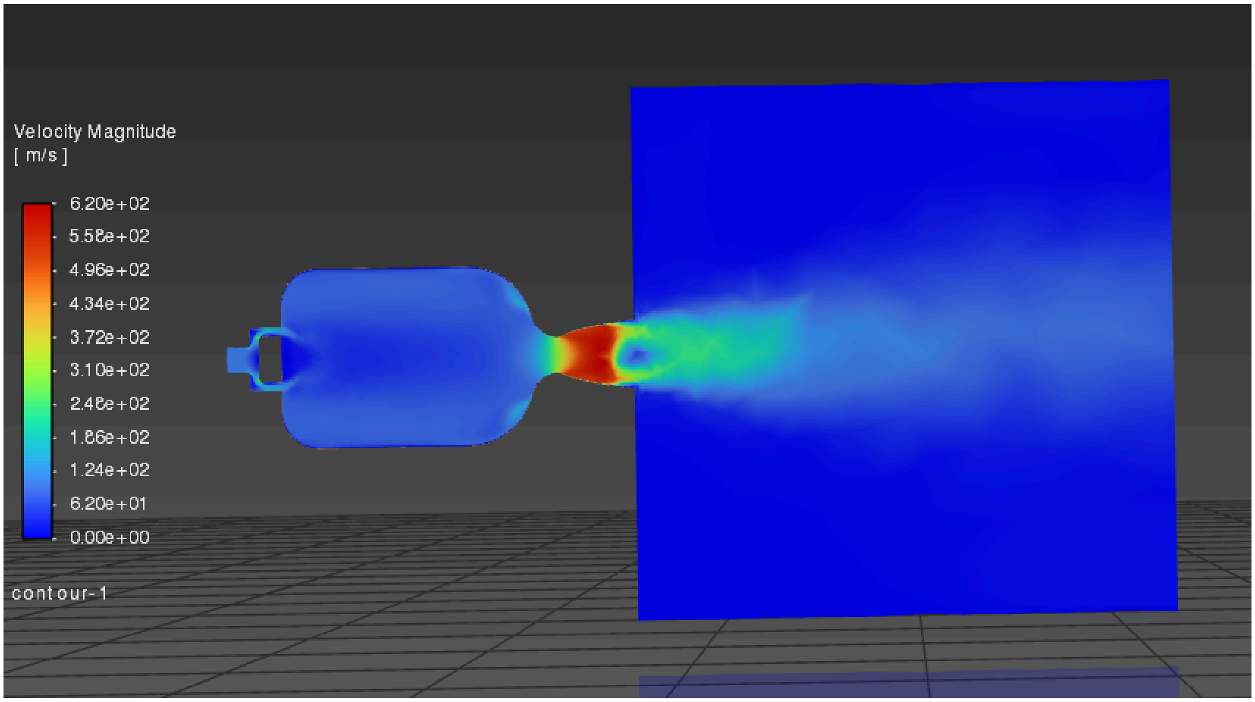


Figure 32: velocity magnitude contour

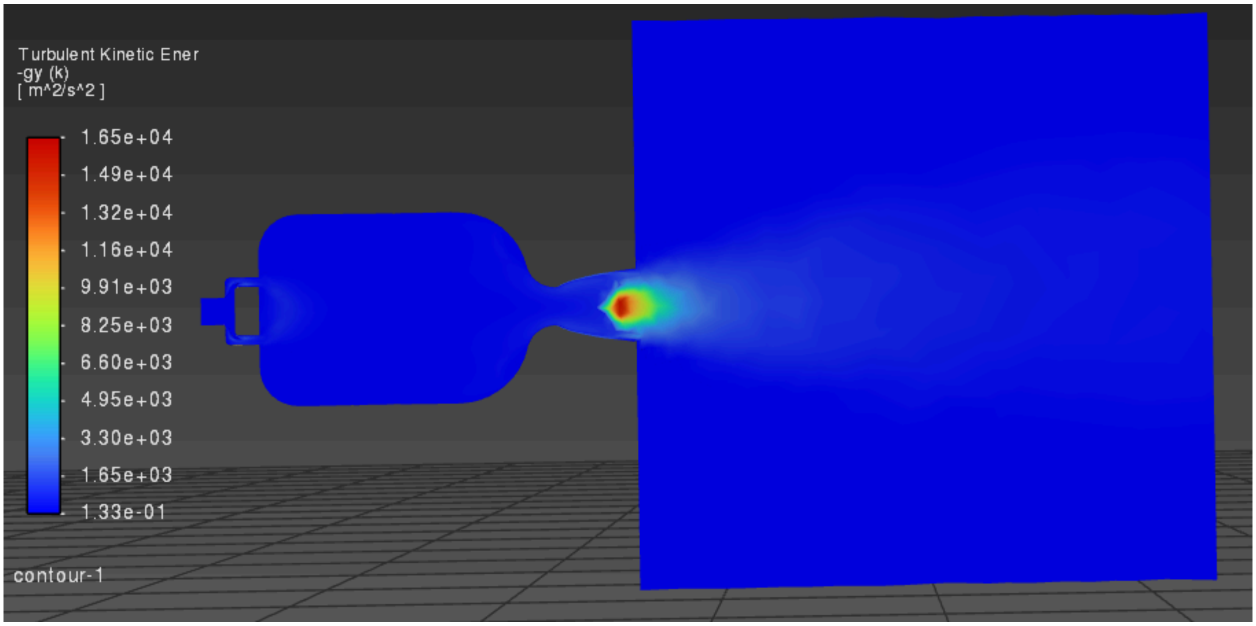


Figure 33: TKE contour

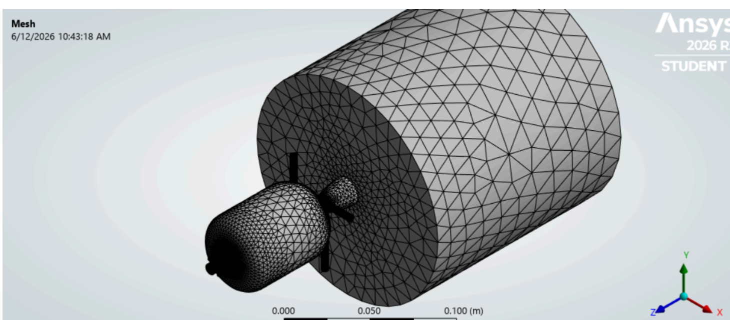
Control: ~53 kPa max chamber pressure, 32 m/s outlet axial velocity, 124 m^2/s^2 average TKE.

The velocity match configuration was designed to reduce the velocity difference between the central fuel jet and the surrounding oxidizer vortex by increasing the oxidizer inlet radius while leaving the fuel inlet unchanged. The objective of this configuration was to create a smoother interaction between the two flow regions, reduce excessive shear, and evaluate whether matching the velocity could improve flow stability and mixing behaviors.

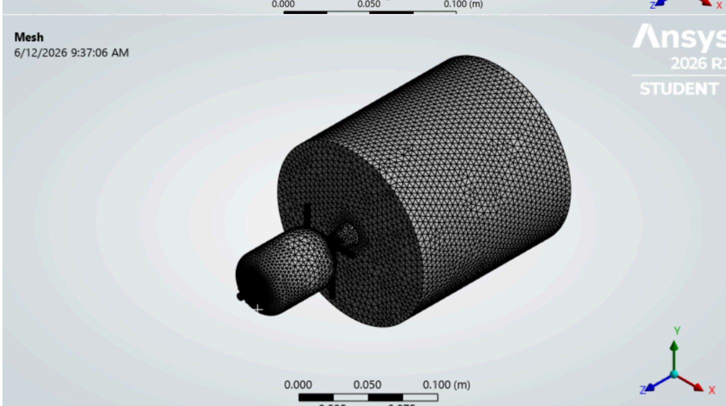
The average TKE converged to 130 $m^2/s^{2,}$ which is a considerable amount greater than the control. This result is reminiscent of the core punch scenario, but the intent of the simulation was vastly different. Matching velocities intuitively would mean a smoother integration between the jet and vortex at the top of the chamber, but it seems that matching velocities may not be the most effective way to do so. It might be that the vortex needs to be fast enough when spiraling up the chamber to not mix with the jet while also remaining slow enough as to not cause too much effective eddy viscosity. Then, at the top, the vortex must slow down enough to match the jet velocity to encourage appropriate mixing. This can be achieved with a different top-of-chamber design or by changing the aspect ratio of the cylinder.

Grid Refinement

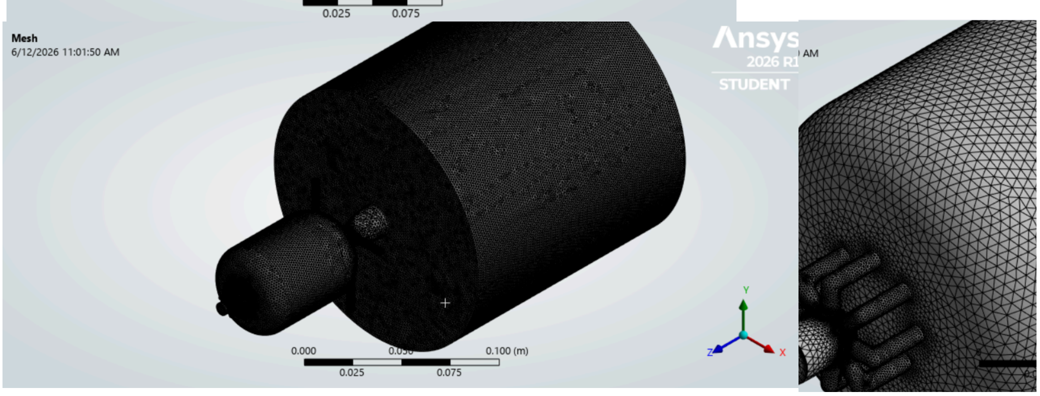
	Run Time	Number of Cells	Oxidizer Inlet Velocity - *Mass Weighted Average*	Fuel Inlet Velocity - *Mass Weighted Average*	Plenum Outlet Velocity - *Mass Weighted Average*	Turbulence Kinetic Energy - *Volume Average*	Pressure - *Volume Average*
Level 1	7:16 min	406,614 Cells	149.47m/s	113.25m/s	36.36m/s	115.60 m^2/s^2	14,221.42Pa
Level 2	15:32 min	590,988 Cells	175.33m/s	144.44m/s	51.31m/s	136.28 m^2/s^2	15,389.54Pa
Level 3	25:32 min	940,029 Cells	153.73m/s	113.66m/s	52.33m/s	135.36 m^2/s^2	14,359Pa



Level 1: 0.5mm element size applied to five faces



Level 2: 5mm element size applied to the whole body



Level 3: 3.5mm element size applied to the whole body

A grid refinement study was conducted to analyze and evaluate the CFD sensitivity to the mesh's resolution. We generated and analyzed three levels and compared each of them. The results of the mass average weight and volume average can be seen in the table above. Level 1 utilized a face sizing of 0.5mm applied only to the five inlet faces, resulting in 406,614 cells. Level 2 applied a body sizing of 5mm to the entire domain, and level 3 further refined the mesh with the same body at a sizing of 3.5mm, which results in the finest mesh analyzed in the results. Level 1 used local face refinement while levels 2 and 3 used body sizing. For each mesh, the oxidizer inlet velocity, fuel inlet velocity, plenum outlet velocity, TKE, and pressure were recorded and compared to determine whether the solution approached mesh independence.

The largest changes occur between level 1 and level 2, where the averaged plenum outlet velocity increases from 36.36m/s to 51.31m/s and the average TKE increases from 115.60m^2/s^2 to 136.28m^2/s^2. This indicates that the coarser mesh was unable to fully resolve the structure of the vortex and turbulence inside the chamber. However, it can be seen from levels 2 to 3 that higher mesh levels produced minor changes in the output quantities. The plenum outlet velocity decreased by 2%, and the average TKE by less than 1%. These results suggest that the solution is approaching grid independence as the mesh is further increased, and mesh density diminishes the changes in the predicted flow field. Given the significant increase in computational costs between levels 1 to 3, Level 2 is considered the most sufficiently refined.

Conclusion

This study investigated the effects of modifying the fuel and oxidizer inlet geometries on the flow characteristics of the Maelstrom vortex-cooled rocket engine. Across all five configurations, the chamber pressure and outlet velocity remain relatively constant, which indicates that the overall flow structure was preserved despite the geometric changes that were applied ot the inlets. The most noticeable differences observed occurred in the core punch and velocity match configurations, where higher TKE values were reported in comparison to the control. These results suggest that altering the velocity relationship between the central fuel jet and the surrounding oxidizer vortex primarily influences local mixing and turbulence rather than the pressure and outlet velocity.

From a pure results standpoint, this investigation demonstrates that modifying performance metrics in highly coupled systems, such as a vortex-cooled rocket engine, requires precise, premeditated geometric adjustments. Even well-reasoned design changes can be counteracted or compromised by overlooked fluid behaviors, emphasizing the difficulty of fluid optimization. This is because changes to one aspect of the flow often influence multiple interacting physical parameters at the same time.

Several limitations should be considered when interpreting the results generated in this report. The simulations were performed as steady-state cold-flow analyses using air, therefore, they do not capture the effects of combustion, heat release, and the time dependence of vortex behavior as the engine is in operation. Additionally, the modifications that were done and analyzed were very small, which likely contributed to the modest quantitative differences between the modified scenarios and the control. Future work should look into larger injector modifications and possibly alternative geometry configurations. A working experimental prototype would also strengthen and validate future work, as a physical product may help to verify trends observed in the CFD results. This would further help to decide whether or not these geometries translate to real-world engine operations.